Classical and Bayesian estimation of multicomponent stress-strength reliability for the Weibull distribution

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Abstract

We study a system that works when at least three of five component strengths exceed a common random stress. Strength and stress follow Weibull distributions with the same shape and different rates. Strength observations are collected in blocks under progressive hybrid censoring, and stress observations are upper k-record values. We derive the reliability function and estimate it by unrestricted and order-restricted maximum likelihood. We also obtain Bayesian estimates under a dependent gamma-Dirichlet prior. A Monte Carlo study with 48 scenarios and 1000 replications per scenario compares absolute bias and mean squared error. Asymptotic, equal-tail and numerical highest posterior density intervals are compared through coverage and average width. An incorrect order restriction increases bias and can prevent intervals from covering the true reliability. Published jute-fibre and carbon-fibre measurements illustrate the methods. The jute example has a Weibull-fit limitation. A random carbon split is used as a boundary check, and separate primary carbon gauge groups provide a further laboratory comparison. Additional studies examine bootstrap intervals, another system size, unequal block designs and prior sensitivity.

Keywords: Weibull distribution; stress-strength reliability; progressive hybrid censoring; upper k-records; maximum likelihood; gamma-Dirichlet prior.

1 Introduction

Reliability is the probability that a system performs its intended task. In a stress-strength model, a component works when its strength is greater than the stress applied to it. A system with several components can continue to work even if some components fail. Here we consider an s-out-of-K system: at least s of the K strengths must exceed the same stress.

Complete observations are not always available in a life test. Some specimens may be withdrawn before failure and the experiment may end at a time limit. Progressive hybrid censoring describes these two features. A block represents a separate test facility. For the stress sample, we retain upper k-records instead of every observation. These records are dependent, so their likelihood differs from the likelihood of independent Weibull observations.

Sultana et al. [1] study block Type-I hybrid censored Weibull populations, including order-restricted inference and a dependent prior. Newer [2] combines block adaptive progressive hybrid censoring with upper k-record sampling for another distribution. Newer and Alanazi [3] discuss multicomponent reliability, Bayesian estimation and bootstrap inference under a different sampling design. We use these papers to develop the assigned Weibull model. We do not use their other distributions or ranked-set likelihoods in place of the model below.

The order assumption in this project is that the strength population is stochastically larger than the stress population. It can improve estimation when it is correct, but it can also produce misleading results when it is false. We therefore include correctly ordered, boundary and incorrectly ordered

cases in the simulation. Sections 2 and 3 present the classical and Bayesian estimates. Section 4 contains the simulation and the data examples. Section 5 gives the conclusions.

1.1 Project outline and notation

We derive the reliability and likelihood, fit unrestricted and ordered models, compare their errors in simulation, and examine published measurements. The raw measurements, retrospective retained observations and simulated data are saved separately. Population truths are known only in simulation; observed-data MSE is therefore not reported for the applications.

Table 1: Main notation. Record order k differs from component count K.

Symbol	Meaning
α	Common Weibull shape
a, b	Strength and stress rates
K, s	Component count and minimum number surviving
k, r	Stress record order and prescribed record count
n_i, m_i, T_i	Block size, target failures and time threshold
d_i, r_{ij}, q_i	Observed failures, actual withdrawals and terminal survivors
d, q	Total strength failures and $q = d + r$
$A(\alpha), B(\alpha)$	Likelihood exposure terms
R, M	Reliability and simulation replications per setting

1.2 Weibull model and system reliability

Let $X_1, \ldots, X_K$ be independent strengths, and let Y be a stress independent of them. All strengths have rate a and stress has rate b, with common shape $\alpha > 0$. For a positive Weibull variable with rate $\lambda > 0$,

$$f(z; \alpha, \lambda) = \alpha\lambda z^{\alpha-1} e^{-\lambda z^\alpha}, \qquad z > 0, \tag{1}$$

$$F(z) = 1 - e^{-\lambda z^\alpha}, \qquad S(z) = e^{-\lambda z^\alpha}, \tag{2}$$

$$h(z) = \alpha\lambda z^{\alpha-1}, \qquad F^{-1}(u) = \left[\frac{-\log(1-u)}{\lambda}\right]^{1/\alpha}. \tag{3}$$

The usual scale is $\lambda^{-1/\alpha}$. We call λ a rate to make the order assumption clear: $a \le b$ means that strength has greater survival than stress at every positive threshold. The shapes below and above one give decreasing and increasing hazards, respectively.

Conditional on $Y = y$, the number of surviving components is binomial with probability e^{-ay^α}. Thus

$$R_{s,K} = \int_0^\infty \sum_{j=s}^{K} \binom{K}{j} e^{-jay^\alpha} (1 - e^{-ay^\alpha})^{K-j} f_Y(y)\, dy. \tag{4}$$

Put $\rho = b/a$ and $u = e^{-ay^\alpha}$. Since $f_Y(y)dy = -\rho u^{\rho-1} du$, integration gives

$$R_{s,K} = \rho \sum_{j=s}^{K} \binom{K}{j} B(\rho + j, K - j + 1). \tag{5}$$

An equivalent expression is easier to evaluate. The transformed strengths X_i^α are exponential with rate a. The $(K - s + 1)$-th smallest transformed strength is a sum of independent exponentials

with rates $Ka, (K-1)a, \ldots, sa$. Taking its Laplace transform at b gives

$$\boxed{R_{s,K} = 1 - \prod_{j=s}^{K} \frac{j}{j+\rho}.} \tag{6}$$

For $K = s = 1$ this reduces to $b/(a+b)$. Reliability tends to zero as $\rho \to 0$ and to one as $\rho \to \infty$. It does not depend directly on the common shape, although the fitted rates do. The main study uses $s = 3$ and $K = 5$; additional checks use other system sizes. In this case $R_{3,5} = 1/2$ at $a = b$, and the order restriction implies $R_{3,5} \geq 1/2$.

The common stress in (4) matters. Component survival indicators are independent conditional on Y, but dependent after averaging over Y. Substituting $P(X > Y)$ directly into an ordinary binomial probability would describe a different experiment. For example, when $s = 2$, $K = 4$ and $\rho = 2$, the shared-stress probability is 0.8000. The single-component probability is $2/3$, but inserting it into an independent binomial calculation gives 0.8889. Those are different experiments. The notebook checks (6) by numerical integration and direct simulation.

More generally, at equal rates the product telescopes to $s/(K+1)$. Therefore the ordered model has lower reliability bound

$$R_{s,K} \geq 1 - \frac{s}{K+1}. \tag{7}$$

This is 0.6 for a 2-out-of-4 system. The implementation uses s and K in the reliability derivative, posterior transformation, HPD support and sampling checks; it does not fix this bound at one half.

1.2.1 Reliability graphs

A larger stress rate makes large stresses less likely, so reliability increases with b/a. Requiring more surviving components reduces reliability. Shared stress creates dependence between the survival indicators.

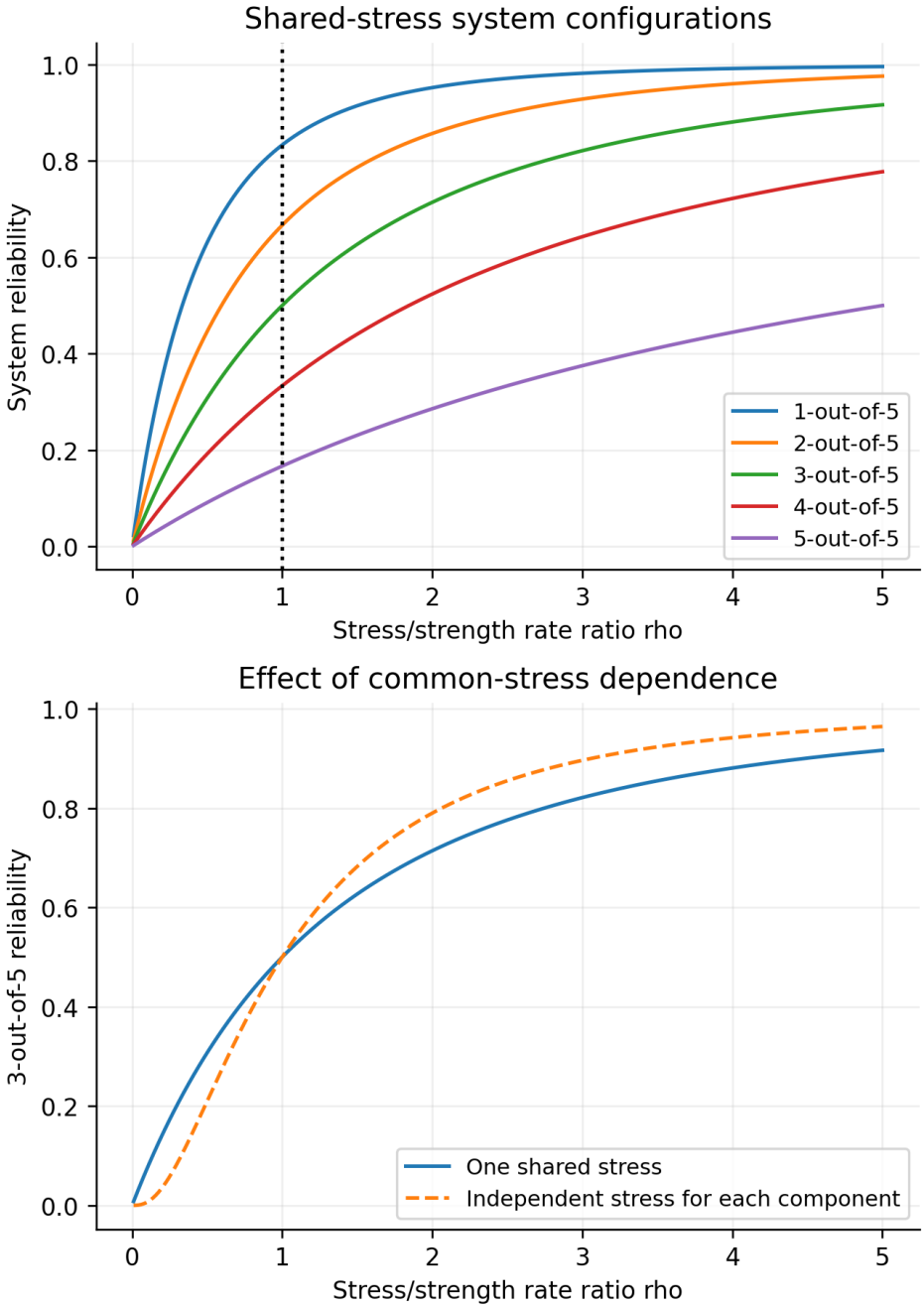


Figure 1: System reliability for different survival requirements, and comparison of shared and independent stresses.

1.3 Observation schemes and likelihood

For strength block i, start with n_i specimens. Choose a target number of failures m_i, a time T_i, and planned withdrawals R_{ij} satisfying $\sum_j R_{ij} = n_i - m_i$. Withdraw survivors uniformly, independently of their unobserved failure times.

The principal Type-I scheme stops at $\tau_i = \min(x_{i,m_i}, T_i)$. Let d_i be the observed number of failures and r_{ij} the actual withdrawals. If the target is reached, all remaining survivors are counted among the final withdrawals. Otherwise the remaining $q_i = n_i - d_i - \sum_j r_{ij}$ specimens are censored at T_i. Up to parameter-free design factors,

$$L_i \propto \prod_{j=1}^{d_i} f_X(x_{ij}) S_X(x_{ij})^{r_{ij}} S_X(\tau_i)^{q_i}. \tag{8}$$

There is no double counting: $q_i = 0$ when final survivors have already been withdrawn. Blocks with no failures still contribute their survival factors. The adaptive Type-II alternative guarantees m_i failures. Withdrawals after T_i are cancelled until all survivors are removed at the target failure. The same likelihood applies using the actual withdrawals. This experiment can continue beyond T_i.

For stress, the first upper k-record is the minimum of the first k observations. Thereafter retain a new record whenever the k-th largest observed value increases. Stop after r distinct records $y_1 < \cdots < y_r$. On the cumulative-hazard scale, successive record increments are independent exponential variables with rate k. The joint record-value likelihood is therefore

$$L_Y = k^r S_Y(y_r)^k \prod_{j=1}^{r} h_Y(y_j) \propto \alpha^r b^r \exp\left[(\alpha - 1)\sum_j \log y_j - bky_r^\alpha\right]. \tag{9}$$

We use this marginal record-value experiment. Acquisition positions are saved to explain extraction, but their waiting times are not included in (9).

Write $d = \sum_i d_i$, $q = d + r$, and define

$$A(\alpha) = \sum_{i,j}(1 + r_{ij})x_{ij}^\alpha + \sum_i q_i \tau_i^\alpha, \tag{10}$$

$$B(\alpha) = ky_r^\alpha, \qquad C = \sum_{i,j} \log x_{ij} + \sum_j \log y_j. \tag{11}$$

The joint log-likelihood, apart from constants, is

$$\ell(\alpha, a, b) = q \log \alpha + d \log a + r \log b + (\alpha - 1)C - aA(\alpha) - bB(\alpha). \tag{12}$$

All calculations below use (12), with positive parameters. The numerical experiments have $d > 0$ and at least two distinct stress records.

2 Maximum Likelihood Estimation

2.1 MLE under no order restriction

For fixed α, differentiation of (12) with respect to the two rates gives

$$\widehat{a}(\alpha) = \frac{d}{A(\alpha)}, \qquad \widehat{b}(\alpha) = \frac{r}{B(\alpha)}. \tag{13}$$

Substitution gives the profile log-likelihood

$$\ell_p(\alpha) = q \log \alpha - d \log A(\alpha) - r \log B(\alpha) + (\alpha - 1)C + \text{constant}. \tag{14}$$

We maximize it over positive α and then use (13). The profile score is

$$\ell_p'(\alpha) = \frac{q}{\alpha} + C - d\frac{A'(\alpha)}{A(\alpha)} - r\frac{B'(\alpha)}{B(\alpha)}. \tag{15}$$

To see why the maximum is unique, write $A(\alpha) = \sum_l w_l z_l^\alpha$. The second derivative of $\log A$ is a weighted variance of $\log z_l$ and is nonnegative. Also $\log B$ is linear in α. Hence $\ell_p''(\alpha) < 0$. Distinct records make the score eventually negative; it is positive near zero. The estimate is a finite interior maximum under the stated conditions. The reliability estimate is $\widehat{R} = R(\widehat{a}, \widehat{b})$.

2.2 MLE under order restriction

Under $a \le b$, keep the rates in (13) if they satisfy the order. Otherwise the maximum is on $a = b$. Differentiating on this boundary gives the pooled rate $q/(A + B)$. Thus, at every trial shape,

$$(\widetilde{a}, \widetilde{b}) = \begin{cases} (d/A, r/B), & d/A \le r/B, \\ (q/(A+B), q/(A+B)), & d/A > r/B. \end{cases} \tag{16}$$

We maximize the resulting profile again over α and calculate $\widetilde{R} = R(\widetilde{a}, \widetilde{b})$. Sorting the two unrestricted rates would not give the correct estimate because strength and stress play different roles. Pooling only at the unrestricted shape would also omit the required reoptimization. On the pooled branch the profile curvature is $-q/\alpha^2 - q\{\log(A + B)\}'' < 0$, and the score agrees across the transition between branches.

2.3 Approximate confidence intervals

With derivatives taken with respect to α, the observed information in the order (α, a, b) is

$$I = \begin{pmatrix} q/\alpha^2 + aA'' + bB'' & A' & B' \\ A' & d/a^2 & 0 \\ B' & 0 & r/b^2 \end{pmatrix}. \tag{17}$$

The derivatives are $A' = \sum_l w_l z_l^\alpha \log z_l$, $A'' = \sum_l w_l z_l^\alpha (\log z_l)^2$, $B' = B \log y_r$, and $B'' = B(\log y_r)^2$. At the unrestricted MLE let $V = I^{-1}$. A positive approximate 95 percent interval for a parameter θ_j has endpoints

$$\exp\left[\log \widehat{\theta}_j \;\pm\; 1.96 \frac{\sqrt{V_{jj}}}{\widehat{\theta}_j}\right]. \tag{18}$$

For reliability use $\psi = \log(b/a)$ with gradient $g = (0, -1/a, 1/b)^\mathsf{T}$. Its estimated standard error is $\sqrt{g^\mathsf{T} V g}$. Exponentiate the endpoints of the normal interval for ψ to obtain ρ_L, ρ_U, then transform them to $[R(1, \rho_L), R(1, \rho_U)]$. This keeps reliability endpoints in $[0, 1]$.

A pooled ordered MLE lies on a boundary, where an ordinary normal approximation is unsuitable. We use percentile parametric bootstrap intervals for it. Each bootstrap replicate regenerates the full strength blocks, withdrawals and stopping, along with the stress records, and refits the ordered model. These intervals are finite-sample approximations. A separate simulation checks their coverage rather than assuming that it is exactly 95 percent.

3 Bayesian Estimation

3.1 Bayesian estimation under no order restriction

We use the gamma-Dirichlet construction described by Sultana et al. [1]. Put $t = a + b$ and $p = a/(a + b)$. Assign independent priors

$$t \sim \text{Gamma}(a_0, b_0), \quad p \sim \text{Beta}(c_1, c_2), \quad \alpha \sim \text{Gamma}(a_1, b_1), \tag{19}$$

where Gamma uses shape and rate. Since $|\partial(a, b)/\partial(t, p)| = t$, the induced joint rate prior is

$$\pi(a, b) = C_0 e^{-b_0(a+b)}(a + b)^{a_0-c_1-c_2} a^{c_1-1} b^{c_2-1}, \tag{20}$$
$$C_0 = \frac{b_0^{a_0}\Gamma(c_1 + c_2)}{\Gamma(a_0)\Gamma(c_1)\Gamma(c_2)}.$$

We choose $(a_0, b_0, c_1, c_2, a_1, b_1) = (2, 0.5, 2, 2, 2, 1)$ before the numerical experiments. This is a proper prior, not a noninformative prior. It gives independent rates only in the special case $a_0 = c_1 + c_2$, which is not our choice. If $c = c_1 + c_2$,

$$\text{Cov}(a, b) = \frac{a_0 c_1 c_2 (c - a_0)}{b_0^2 c^2 (c + 1)} = 0.8. \tag{21}$$

The prior means of a, b and α are all two. The same hyperparameters are used for every simulation scenario; they are not fitted to the true simulation rates.

Multiplying the likelihood and prior, and including the transformation Jacobian, gives

$$\pi(\alpha, t, p \mid D) \propto \alpha^{q+a_1-1} e^{(\alpha-1)C-b_1\alpha} p^{d+c_1-1}(1 - p)^{r+c_2-1}$$
$$\times\, t^{q+a_0-1} e^{-tH(\alpha,p)}, \tag{22}$$
$$H(\alpha, p) = b_0 + pA(\alpha) + (1 - p)B(\alpha).$$

Therefore $t \mid \alpha, p, D \sim \text{Gamma}(q + a_0, H)$. Integrating out t gives

$$\pi(\alpha, p \mid D) \propto \alpha^{q+a_1-1} e^{(\alpha-1)C-b_1\alpha} p^{d+c_1-1}(1 - p)^{r+c_2-1} H^{-(q+a_0)}. \tag{23}$$

We integrate (23) numerically over $\alpha > 0$ and $0 < p < 1$. Specifically, Gauss-Legendre quadrature is applied to $\log \alpha \in [\log(0.02), \log(30)]$ and p on its support, including the log-shape Jacobian. Small edge weights and finer-grid comparisons check the numerical bounds. Conditional expectations of t give

$$E(a \mid D) = E\left[\frac{p(q + a_0)}{H} \mid D\right], \qquad E(b \mid D) = E\left[\frac{(1 - p)(q + a_0)}{H} \mid D\right]. \tag{24}$$

Under squared error loss the reliability estimate is $\widehat{R}_B = E[R(p, 1-p) \mid D]$, not reliability evaluated at the posterior mean rates. The notebook also reports LINEX estimates. Under loss $L(\delta, R) = e^{c(\delta-R)} - c(\delta - R) - 1$, differentiation of posterior expected loss gives

$$\widehat{R}_{B,c} = -\frac{1}{c} \log E(e^{-cR} \mid D), \qquad c \neq 0. \tag{25}$$

We use $c = 1.5$ and $c = -1.5$ as supplementary calculations.

3.2 Bayesian estimation under order restriction

The restriction $a \leq b$ is equivalent to $p \leq 1/2$. We condition the joint prior on this event:

$$\pi_O(\alpha, a, b) = \frac{\pi(\alpha, a, b)\, \mathbf{1}\{a \leq b\}}{I_{1/2}(c_1, c_2)}. \tag{26}$$

For the symmetric Beta prior used here, the denominator is $1/2$. The posterior kernel is still (23), but its support is $\alpha > 0$, $0 < p \leq 1/2$. We compute posterior means on that support. We do not obtain the ordered posterior by sorting draws from the unrestricted posterior.

3.3 Credible intervals and numerical checks

An equal-tail interval uses the posterior 0.025 and 0.975 quantiles of R. A highest posterior density (HPD) set contains points with density above a threshold and total posterior probability 0.95. For a unimodal marginal it is a shortest interval of that probability.

For a numerical HPD interval, first transform the marginal density of p to the reliability scale using $f_R(R(p)) = f_p(p)/|R'(p)|$. Interpolate the density with a shape-preserving cubic function, integrate it, and minimize the length of an interval containing probability 0.95. Both ends of the support are included as candidate endpoints. We check probability mass against 0.95 and compare endpoints on a finer quadrature grid. These are numerical HPD approximations; they are not exact closed-form intervals. If the density were multimodal, a single shortest interval would not necessarily be the full HPD set.

The distinction matters at an order boundary. A continuous ordered equal-tail interval starts above 1/2, so it cannot cover a true reliability of exactly 1/2. An ordered HPD interval can start at 1/2. Under a false restriction the true reliability is below 1/2, outside the support of both interval constructions.

The simulation uses 80×110 quadrature nodes and the data examples use 240×320. Application results are checked using 400×520 nodes. As a further check, the notebook compares quadrature with four Metropolis-Hastings chains and reports traces, effective sample sizes and split R-hat. These are numerical checks on the posterior calculations, not extra estimation methods in the main comparison.

4 Numerical Illustration

4.1 Simulation study

There are 48 scenarios: three rate-order regimes, two shapes, two block sizes, two censoring protocols and two record orders. The rate pairs are $(0.8, 1.6)$, $(1, 1)$ and $(1.6, 0.8)$, called ordered, boundary and violated. Shapes are 0.8 and 1.8. Each experiment has three strength blocks of size 20 or 40, target $m = \lfloor 0.7n \rfloor$, and balanced planned withdrawals. The time limit is fixed at the true strength distribution's 65th percentile before generating the data. Both Type-I and adaptive Type-II progressive hybrid protocols are considered. The record order is $k = 1$ or $k = 3$, with six records for $n = 20$ and twelve for $n = 40$.

Each scenario has $M = 1000$ independent replications, giving 48,000 datasets. All four estimators use the same data within a replication. Increasing the size also increases the number of stress records, so it does not isolate the effect of strength sample size. A fixed master seed of 250150045 makes the calculations reproducible.

If $e_l = \widehat{\theta}_l - \theta$ is the error in replication l, we calculate

$$\mathrm{AB} = \left| \frac{1}{M} \sum_{l=1}^{M} e_l \right|, \qquad \mathrm{MSE} = \frac{1}{M} \sum_{l=1}^{M} e_l^2. \tag{27}$$

Here AB is absolute bias, not mean absolute error. For intervals $[L_l, U_l]$, coverage probability (CP) and average width (AW) are

$$\widehat{\mathrm{CP}} = \frac{1}{M} \sum_{l=1}^{M} \mathbf{1}\{L_l \le \theta \le U_l\}, \qquad \mathrm{AW} = \frac{1}{M} \sum_{l=1}^{M} (U_l - L_l). \tag{28}$$

The estimated Monte Carlo standard errors are

$$\mathrm{MCSE}(\mathrm{Bias}) = \frac{\mathrm{SD}(e_l)}{\sqrt{M}}, \tag{29}$$

$$\mathrm{MCSE}(\widehat{\mathrm{CP}}) = \sqrt{\frac{\widehat{\mathrm{CP}}(1-\widehat{\mathrm{CP}})}{M}}. \tag{30}$$

They are saved with the scenario-level results. The notebook and CSV tables also contain parameter errors for α, a, b, as well as RMSE. The manuscript tables focus on the assigned system reliability.

4.1.1 Simulation algorithms

Strength lifetimes are generated completely, but only observed failures and censoring information enter estimation. Withdrawn lifetimes are hidden. For stress records, independent unit-mean exponential variables give

$$y_j = \left(\frac{\sum_{l=1}^{j} E_l}{kb}\right)^{1/\alpha}, \qquad j = 1, \ldots, r. \tag{31}$$

This generates the marginal record values directly, without waiting times. Published-data examples instead scan their stated acquisition sequence.

Algorithm 1 Generate one retained dataset

Input: (α, a, b), block designs, protocol, k and r

for each strength block i **do**
  Generate n_i lifetimes using (3); mark them active
  while fewer than m_i failures have been observed **do**
    Find the next active failure
    if Type I and the next failure exceeds T_i **then**
      Censor active survivors at T_i and stop the block
    else
      Record the failure and remove that specimen
      Withdraw uniformly selected survivors using the planned count
      For adaptive Type II, use zero planned withdrawals after T_i
      At the target failure, withdraw all survivors and stop the block
    end if
  end while
  Save failures, actual withdrawals and terminal survivors
end for
Generate r stress records using (31)
Check specimen conservation and return only retained observations

Posterior integration includes the log-shape Jacobian and quadrature weights. Conditional Gamma means recover the two rate means after integrating out t. The reliability marginal gives equal-tail intervals; transforming its density and finding the shortest 0.95 interval gives the numerical HPD approximation. Finer grids check both calculations. A pooled ordered MLE receives no invented regular-normal interval. Its bootstrap preserves the full observation scheme, as described in Section 4.5.

Algorithm 2 Compare the four estimators

Input: Forty-eight settings, $M = 1000$, fixed prior and seed

for each setting and replication **do**
  Generate one dataset by Algorithm 1
  Form $A(\alpha), B(\alpha), C, d, r$ from the retained data
  Maximize both profiles, including order-boundary pooling
  Integrate (23) on each posterior support
  Save means, equal-tail and numerical HPD intervals
  Save the unrestricted log-ratio ACI and error against the fixed truth
end for
Summarize bias, MSE, coverage, width and their Monte Carlo errors
Assess bootstrap coverage separately in twelve selected settings

Table 2: Absolute bias and MSE of reliability, averaged over 16 scenarios in each rate regime.

Rate regime	Estimator	Absolute bias	MSE
Ordered	MLE	0.0067	0.0132
Ordered	Ordered MLE	0.0060	0.0119
Ordered	Bayes	0.0273	0.0100
Ordered	Ordered Bayes	0.0047	0.0052
Boundary	MLE	0.0085	0.0175
Boundary	Ordered MLE	0.0565	0.0100
Boundary	Bayes	0.0113	0.0117
Boundary	Ordered Bayes	0.1083	0.0143
Violated	MLE	0.0099	0.0133
Violated	Ordered MLE	0.1977	0.0397
Violated	Bayes	0.0376	0.0108
Violated	Ordered Bayes	0.2510	0.0637

4.1.2 Results for the model parameters

Reliability depends on the rate ratio, but the common shape and individual rates are also estimated from the retained observations. The next three tables therefore report their point-estimation errors. These are different targets with different scales. An MSE for a should not be compared numerically with an MSE for R as if both measured the same quantity.

Table 3: Point-estimation errors for α, averaged over the 16 scenarios in each rate regime.

Regime	Estimator	Absolute bias	MSE	RMSE
Ordered	MLE	0.0407	0.0297	0.1552
Ordered	Ordered MLE	0.0390	0.0286	0.1527
Ordered	Bayes	0.0427	0.0271	0.1498
Ordered	Ordered Bayes	0.0334	0.0246	0.1428
Boundary	MLE	0.0440	0.0298	0.1557
Boundary	Ordered MLE	0.0167	0.0220	0.1340
Boundary	Bayes	0.0365	0.0253	0.1449
Boundary	Ordered Bayes	0.0264	0.0204	0.1300
Violated	MLE	0.0460	0.0305	0.1576
Violated	Ordered MLE	0.0936	0.0268	0.1514
Violated	Bayes	0.0249	0.0231	0.1389
Violated	Ordered Bayes	0.1234	0.0340	0.1695

Table 4: Point-estimation errors for a, averaged over the 16 scenarios in each rate regime.

Regime	Estimator	Absolute bias	MSE	RMSE
Ordered	MLE	0.0127	0.0164	0.1252
Ordered	Ordered MLE	0.0117	0.0161	0.1242
Ordered	Bayes	0.0262	0.0166	0.1261
Ordered	Ordered Bayes	0.0174	0.0150	0.1202
Boundary	MLE	0.0230	0.0270	0.1603
Boundary	Ordered MLE	0.0053	0.0213	0.1432
Boundary	Bayes	0.0307	0.0259	0.1575
Boundary	Ordered Bayes	0.0231	0.0194	0.1370
Violated	MLE	0.0651	0.0913	0.2923
Violated	Ordered MLE	0.1787	0.0756	0.2697
Violated	Bayes	0.0426	0.0746	0.2662
Violated	Ordered Bayes	0.2296	0.0918	0.2964

Table 5: Point-estimation errors for b, averaged over the 16 scenarios in each rate regime.

Regime	Estimator	Absolute bias	MSE	RMSE
Ordered	MLE	0.2020	0.7218	0.8167
Ordered	Ordered MLE	0.2075	0.7128	0.8116
Ordered	Bayes	0.1183	0.3682	0.5980
Ordered	Ordered Bayes	0.1911	0.3398	0.5735
Boundary	MLE	0.1193	0.2979	0.5260
Boundary	Ordered MLE	0.2394	0.2632	0.4909
Boundary	Bayes	0.1645	0.2193	0.4579
Boundary	Ordered Bayes	0.4384	0.3129	0.5477
Violated	MLE	0.0854	0.1829	0.4166
Violated	Ordered MLE	0.6502	0.4958	0.7012
Violated	Bayes	0.1844	0.1750	0.4098
Violated	Ordered Bayes	0.8698	0.8401	0.9123

All table entries first summarize the 1000 replications within a scenario and then average over the 16 scenarios in a regime. Thus the root mean squared error column is an average of scenario RMSE values, rather than the square root of the displayed average MSE. Detailed results retain each scenario separately. Averaging is useful for a short comparison, but it can hide variation between shapes, sizes and sampling designs.

The order assumption acts directly on the two rates. When it is false, pooling or truncation changes their estimates and can also change the fitted common shape. A prior may reduce a variance while shifting an estimate away from the truth. This is why the comparison uses MSE as well as bias and avoids calling one method best for every target.

For the ordered rate regime, mean scenario-specific reliability MSE is 0.0132 for unrestricted MLE, 0.0100 for unrestricted Bayes and 0.0052 for ordered Bayes.

For the boundary rate regime, mean scenario-specific reliability MSE is 0.0175 for unrestricted MLE, 0.0117 for unrestricted Bayes and 0.0143 for ordered Bayes.

For the violated rate regime, mean scenario-specific reliability MSE is 0.0133 for unrestricted MLE, 0.0108 for unrestricted Bayes and 0.0637 for ordered Bayes.

4.1.3 Information retained under each protocol

The starting number of specimens is not the observed failure count. Type-I stopping can occur before the target. Adaptive Type II continues until its target is reached. Both experiments include censoring information in their exposure terms, but they do not retain the same failures or have the same duration. The histograms in the simulation figures show this difference directly. Under

Table 6: Retained strength information across the simulation. Means pool all rate and shape settings at the indicated size and protocol. The target is the sum across three blocks.

Protocol	n	Target	Mean failures	SD	Time-stopped blocks
Adaptive II	20	42	42.00	0.00	0.00
Adaptive II	40	84	84.00	0.00	0.00
Type I	20	42	28.98	2.81	2.99
Type I	40	84	57.70	3.98	3.00

adaptive Type II the totals are fixed at 42 or 84 failures because there are three blocks. Under Type I the failure count varies. More observed failures do not by themselves establish that one design is preferable, since test duration and withdrawals also change. Tables 2 and 7 average the 16 scenario-specific summaries within each rate regime. In particular, the reported AB is the average of the scenario-specific absolute biases. It is not the absolute value of an average signed bias.

The results show why an order restriction must be justified. In the ordered regime, the ordered Bayesian estimator has a smaller reliability MSE than the unrestricted methods in this experiment. In the violated regime, the restricted methods cannot estimate reliability below one half and have much larger bias. There is no general advantage to restricting the model when the assumed order is false.

At the equality boundary, ordered equal-tail intervals have coverage 0.000, whereas ordered numerical HPD intervals have coverage 0.909. Their mean widths are 0.254 and 0.233, respectively. The HPD calculation allows the lower endpoint to equal one half. This explains why its boundary coverage differs from the equal-tail result; it does not establish nominal coverage in every scenario.

Table 7: Coverage probability and average width of 95 percent reliability intervals, averaged over 16 scenarios in each regime. HPD denotes the numerical approximation described in Section 3.3.

Rate regime	Interval	CP	AW
Ordered	MLE ACI	0.939	0.400
Ordered	Bayes equal-tail	0.967	0.401
Ordered	Bayes HPD	0.951	0.391
Ordered	Ordered Bayes equal-tail	0.982	0.323
Ordered	Ordered Bayes HPD	0.954	0.313
Boundary	MLE ACI	0.940	0.453
Boundary	Bayes equal-tail	0.963	0.430
Boundary	Bayes HPD	0.942	0.426
Boundary	Ordered Bayes equal-tail	0.000	0.254
Boundary	Ordered Bayes HPD	0.909	0.233
Violated	MLE ACI	0.937	0.407
Violated	Bayes equal-tail	0.957	0.387
Violated	Bayes HPD	0.944	0.381
Violated	Ordered Bayes equal-tail	0.000	0.176
Violated	Ordered Bayes HPD	0.000	0.153

4.1.4 Reading the simulation graphs

Figures 2–4 show how the reliability errors change with the size setting. Each plotted point averages eight scenarios at that size. The upper panel is MSE and the lower panel is signed bias. A point closer to zero in the bias panel does not necessarily have the lowest MSE, because MSE also reflects variation across datasets.

In the ordered cases the order restriction can reduce uncertainty by excluding rate pairs that are assumed impossible. At the boundary, it tends to shift reliability upward. In the violated cases, a restricted reliability estimate cannot approach the true value below one half, even if more observations are collected. The bias in that case is a consequence of the assumption rather than a numerical fitting failure.

The HPD comparison makes the boundary distinction visible. An ordered equal-tail interval starts strictly above one half, whereas a shortest interval can use one half as an endpoint. Shorter numerical HPD intervals therefore need to be assessed by coverage as well as width. The reported averages are not a claim that every scenario attains nominal coverage.

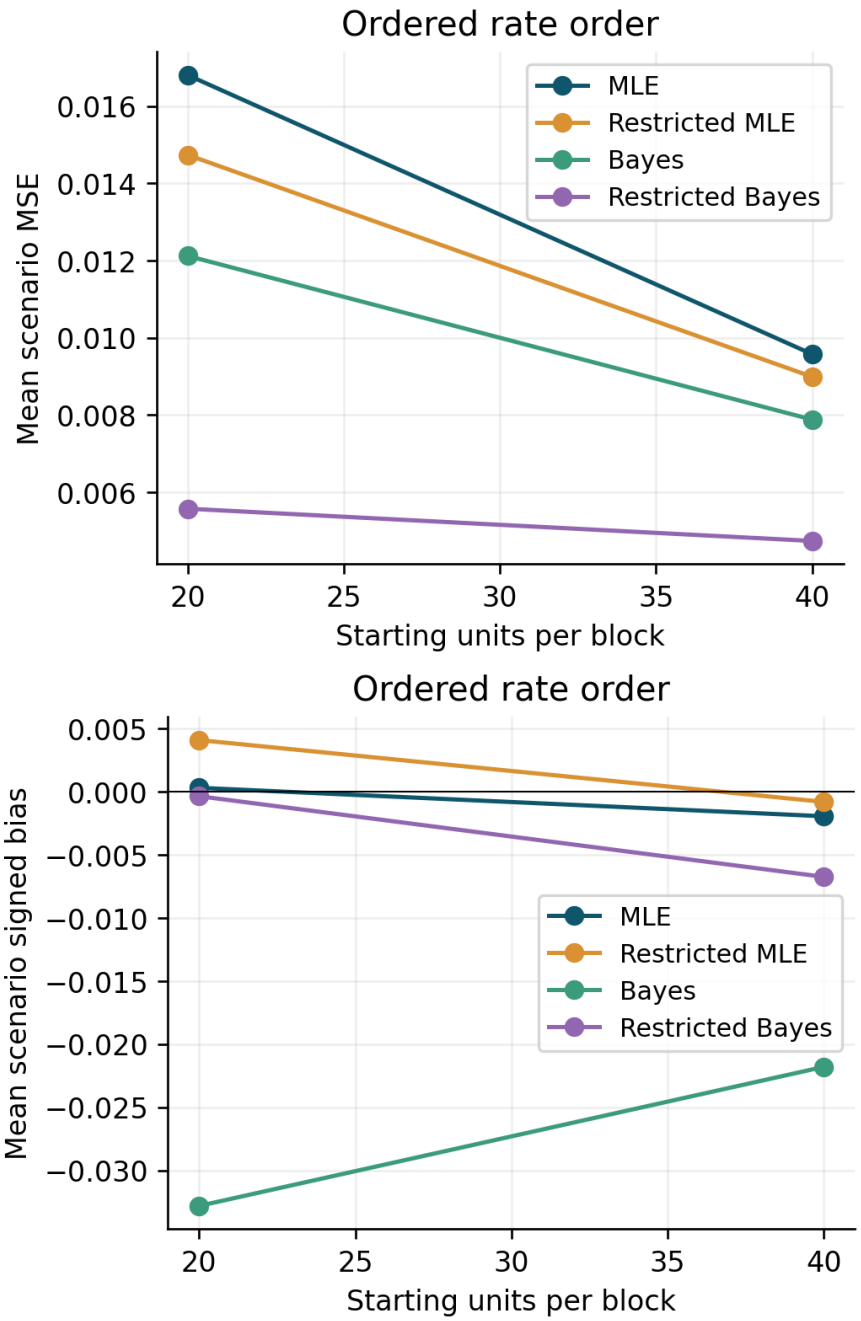


Figure 2: Reliability MSE and signed bias in the ordered rate regime. Both the strength size and stress record count increase between the two horizontal-axis values.

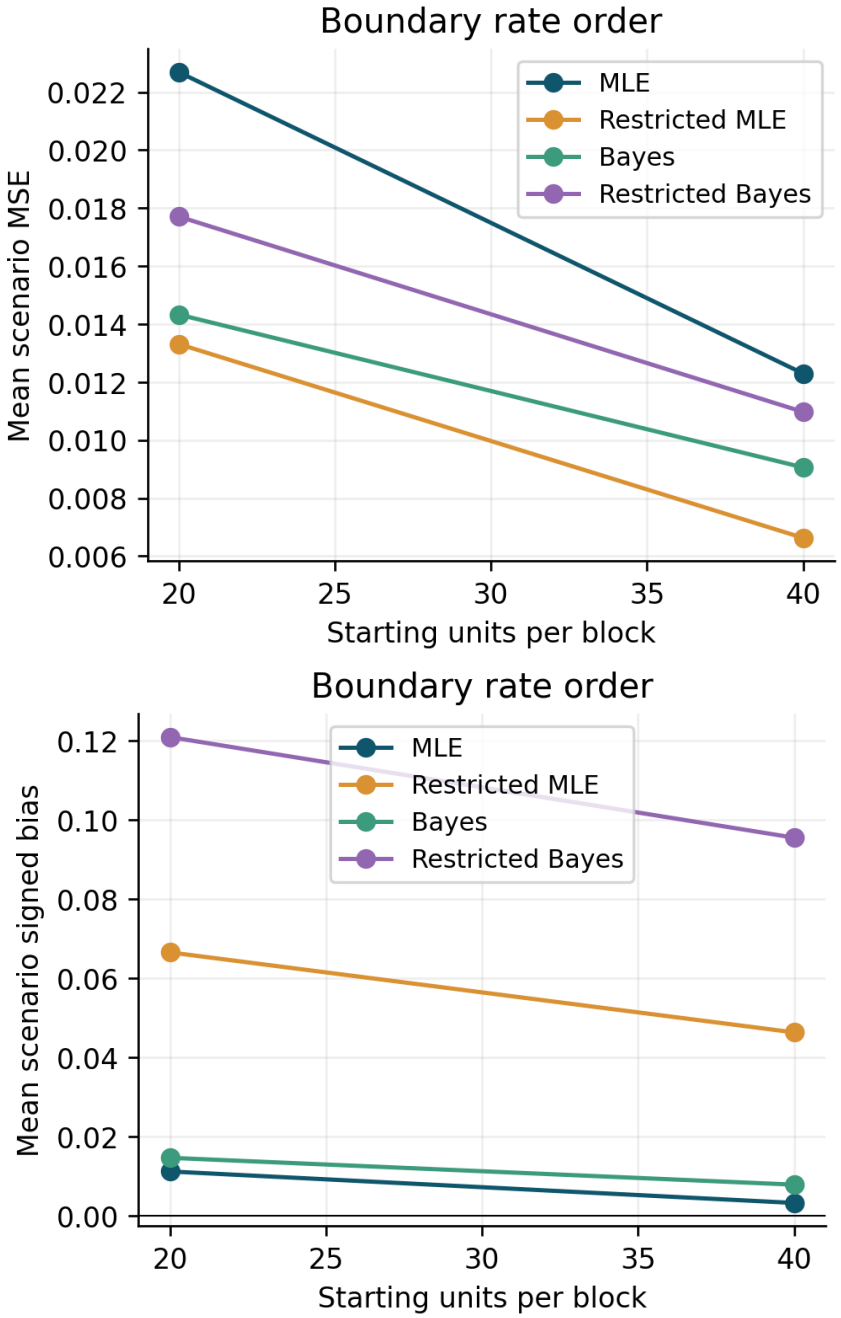


Figure 3: Reliability MSE and signed bias in the boundary rate regime. Both the strength size and stress record count increase between the two horizontal-axis values.

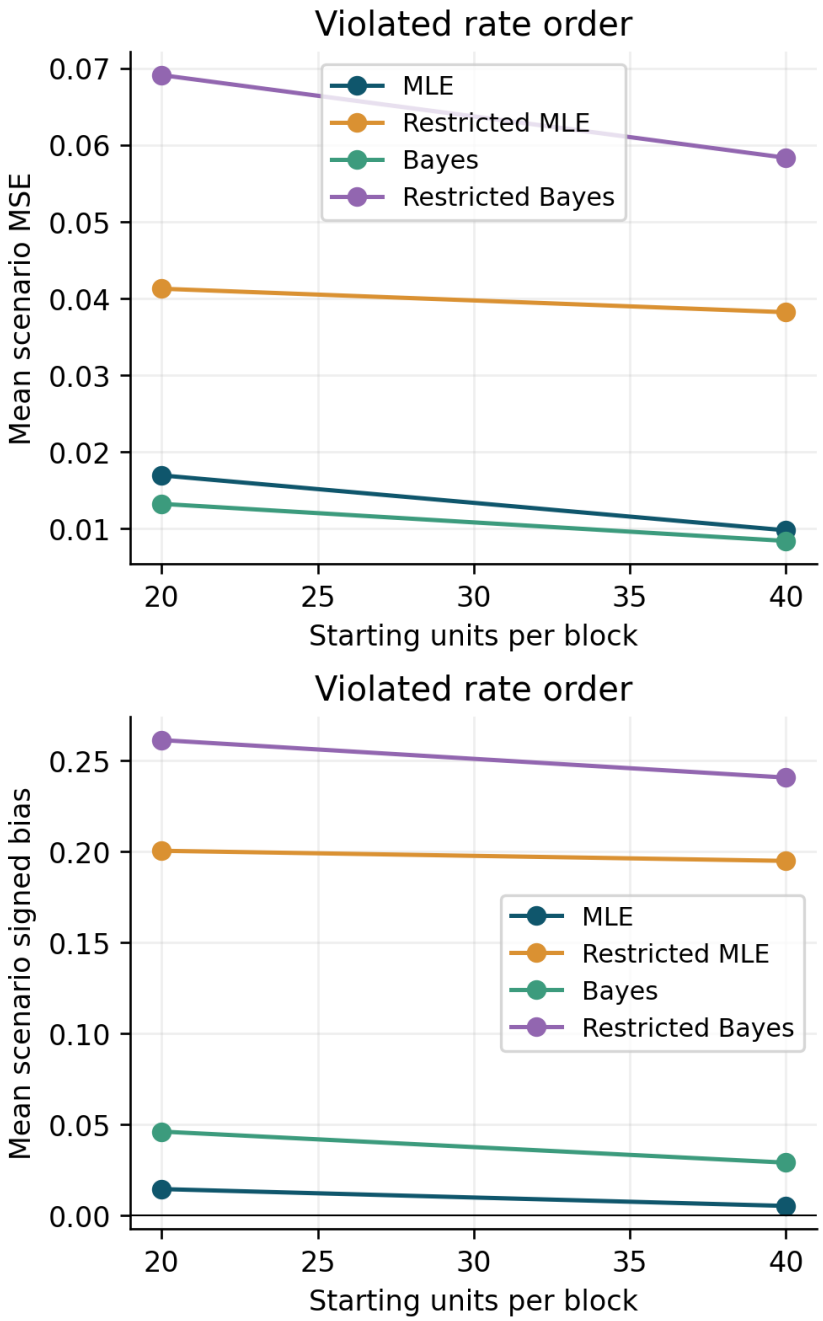


Figure 4: Reliability MSE and signed bias in the violated rate regime. Both the strength size and stress record count increase between the two horizontal-axis values.

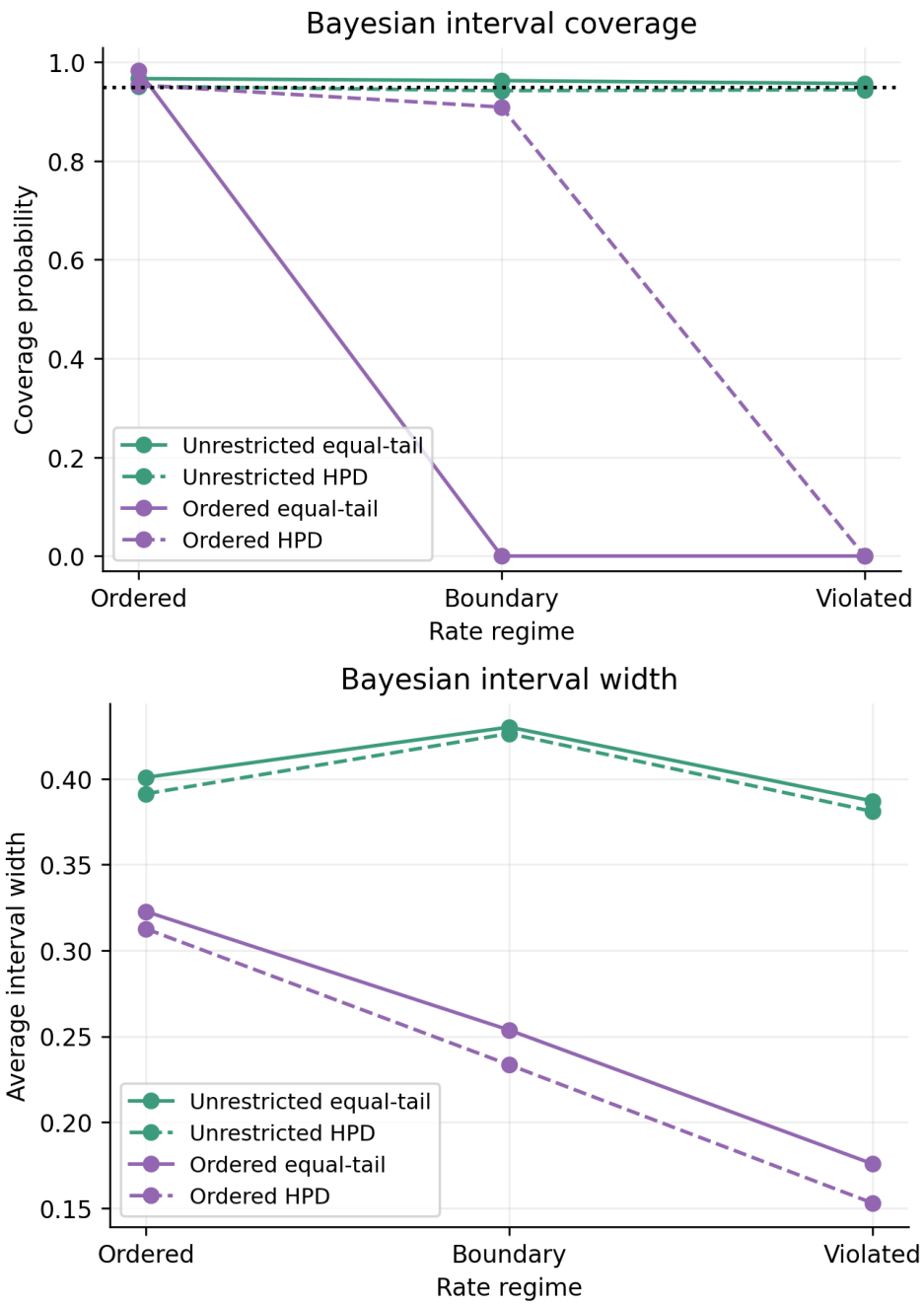


Figure 5: Coverage and average width for unrestricted and ordered Bayesian equal-tail and numerical HPD intervals. Each value averages 16 scenario summaries.

Ordered MLE bootstrap coverage is assessed separately in twelve selected settings, with 500 outer datasets and 399 bootstrap resamples each. These settings cover all three order regimes, both sizes and both protocols at shape 1.8 and record order one. They do not assign a regular normal interval to a boundary estimate. The unrestricted percentile and studentized comparisons appear in Section 4.5.

Table 8: Ordered MLE percentile-bootstrap intervals in 12 settings: 500 outer datasets and 399 resamples per dataset. MCSE refers to coverage.

Regime	n	Protocol	CP	MCSE	AW
Ordered	20	Type I	0.932	0.011	0.381
Ordered	20	Adaptive II	0.940	0.011	0.373
Ordered	40	Type I	0.974	0.007	0.340
Ordered	40	Adaptive II	0.948	0.010	0.316
Boundary	20	Type I	0.958	0.009	0.354
Boundary	20	Adaptive II	0.950	0.010	0.335
Boundary	40	Type I	0.978	0.007	0.279
Boundary	40	Adaptive II	0.962	0.009	0.249
Violated	20	Type I	0.000	0.000	0.327
Violated	20	Adaptive II	0.000	0.000	0.303
Violated	40	Type I	0.000	0.000	0.252
Violated	40	Adaptive II	0.000	0.000	0.216

4.2 Real data analysis

The examples use published breaking-strength measurements. Both strength and comparison stress are measured in the same units within an example. The comparison group is a laboratory reference population, not a set of measured operating loads. The calculated reliability should therefore not be described as a certified in-service probability.

Table 9: Descriptive statistics for all supplied observations used in the applications, before retrospective censoring. Jute values use the source scaling; carbon values in this table are in GPa.

Sample	n	Mean	SD	Median	Minimum	Maximum
Jute 15 mm	30	0.3639	0.2430	0.3474	0.0427	0.8139
Jute 20 mm	30	0.3407	0.2473	0.2646	0.0367	0.7651
Carbon (GPa)	100	2.6214	1.0139	2.7000	0.3900	5.5600

4.2.1 Jute-fibre measurements

The 15 mm and 20 mm jute-fibre groups have 30 observations each. They are transcribed, in printed order, from Table 10 of Sultana et al. [1], where they are attributed to Xia et al. [4]. The reproduced measurements have already been divided by 1000. We use 15 mm as strength and 20 mm as comparison stress. The 10 mm group is excluded because the reproduced table contains the unverified value 70.32383. We do not guess a replacement for it. The original Xia experimental file was not available; the supplied secondary table is the transcription source.

We first fit the complete groups and assess Weibull fit using bootstrap-calibrated Kolmogorov-Smirnov tests with refitting in each of 1000 resamples. We also test the common-shape assumption by a bootstrap likelihood-ratio test. Fitted-parameter K-S tests need this calibration; ordinary distribution-free K-S p-values would not apply. For the jute groups, the common-shape bootstrap

Table 10: Complete-data Weibull fits and bootstrap-calibrated K-S tests. Carbon rates use the fixed 3 GPa measurement unit.

Sample	n	$\widehat{\alpha}$	Rate	K-S	p-value
Jute strength	30	1.499	3.908	0.169	0.026
Jute comparison	30	1.361	3.839	0.149	0.078
Carbon strength	50	2.719	1.180	0.078	0.585
Carbon comparison	50	2.897	0.942	0.113	0.101

p-value is 0.620. The calibrated K-S p-values are 0.026 and 0.078. Thus the first group rejects Weibull adequacy at 5 percent, even though the common-shape test does not reject. The jute estimates must be read with this fit limitation. The carbon common-shape bootstrap p-value is 0.678. Table 10 reports its separate fit checks.

Next, a seeded allocation divides the strength group into three blocks of ten. Each block has target seven failures, balanced withdrawals and $T = 0.5$. Type-I progressive hybrid censoring leaves 13 observed failures altogether. From the comparison group we retain the first six upper 2-records, stopping at position nine. The printed order is an imposed acquisition order; it is not known to be the historical order of the measurements. The notebook examines alternative random orders as a sensitivity check. Allocation, withdrawals, terminal survivors and record positions are saved in the data folder.

4.2.2 Jute fit and retained observations

The Q-Q plots use all complete measurements and must be read with the calibrated fit tests. The rejected fit in the 15 mm group remains a limitation. Density and CDF versions are saved in the notebook and graphs folder. The sampling plots identify which observations enter the likelihood. Ring size shows actual withdrawals; terminal survivors contribute survival information at the time limit. The stress plot distinguishes the complete sequence from the retained upper records.

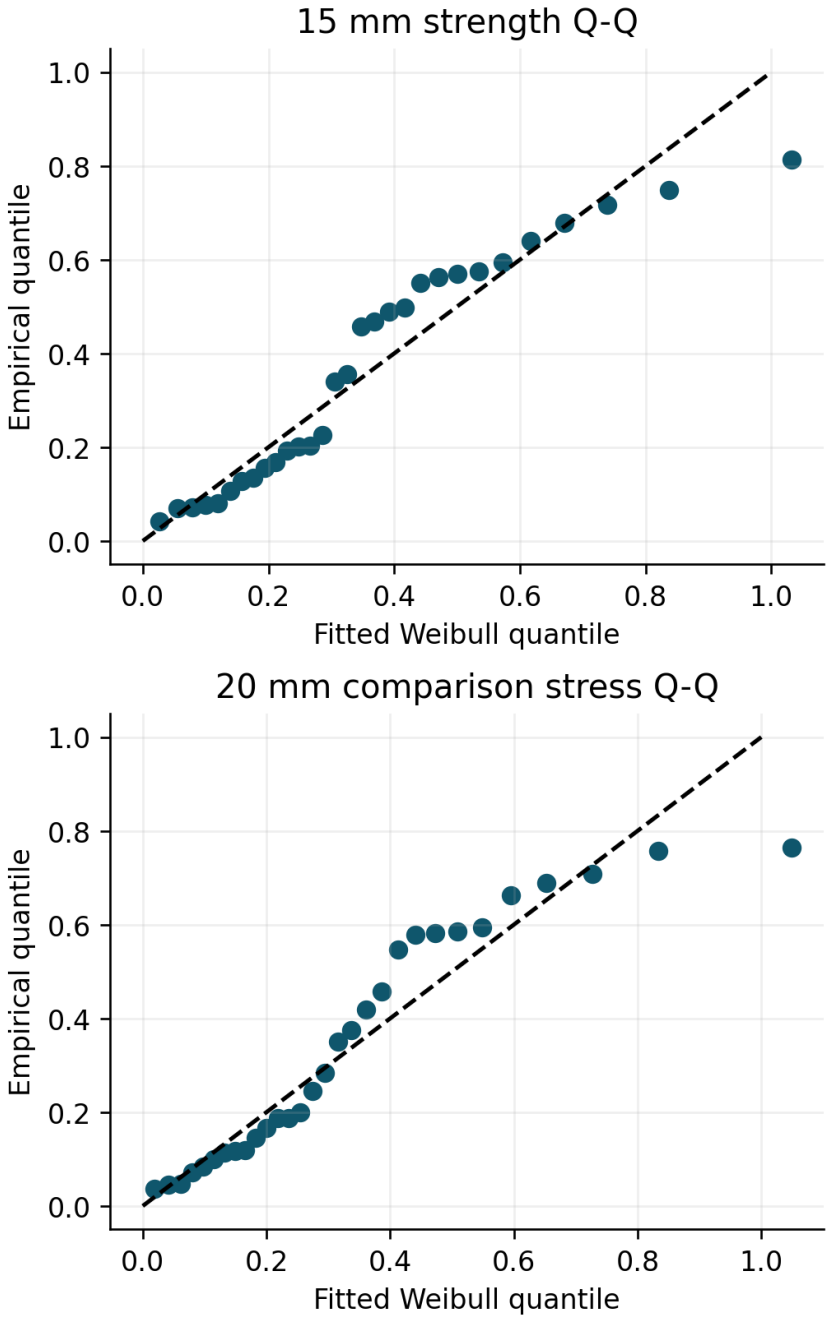


Figure 6: Jute Weibull Q-Q plots for the 15 mm and 20 mm groups.

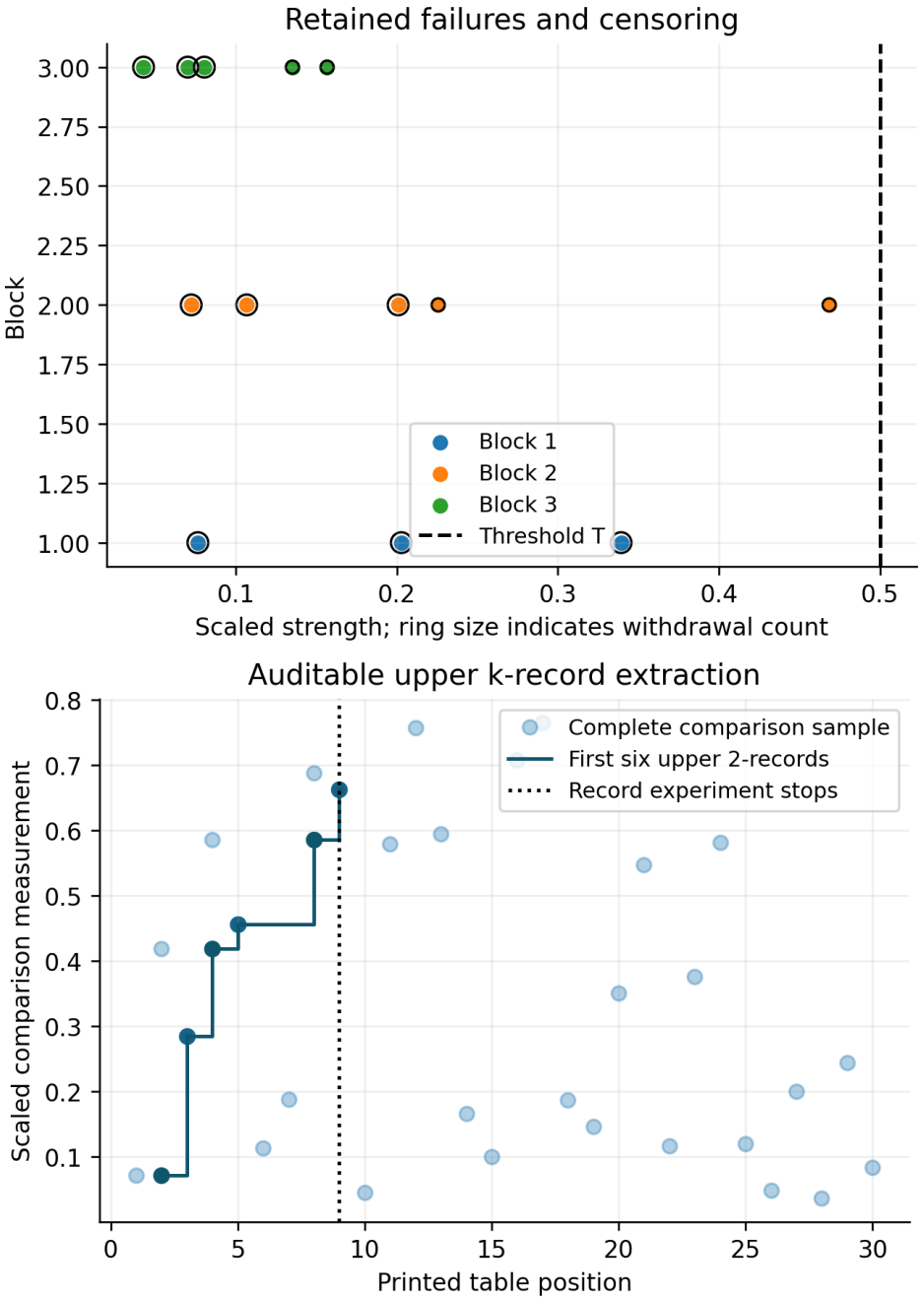


Figure 7: Jute strength censoring and upper 2-record extraction.

4.2.3 Carbon random-split boundary check

As a check near an equal-population boundary, we also use the 100 carbon-fibre measurements reproduced by Newer and Alanazi [3], Section 5.1.1, attributed there to Nichols and Padgett [5]. They are in GPa. A seeded random permutation splits them into disjoint groups of fifty. We divide both groups by a fixed 3 GPa unit before inference, and keep the same prior hyperparameters on these scaled rates. The original Nichols and Padgett experimental file was not supplied; the reproduced article is the source for our transcription.

Strength block sizes are 17, 17 and 16, with target $\lfloor 0.7n \rfloor$, $T = 1$ in scaled units and balanced withdrawals. We use the first five upper 4-records from the comparison half. The retained records are distinct, so no jitter is added to rounded source values. Both complete groups are compatible with the fitted Weibull distributions at the 5 percent level, and the common-shape test does not reject. The two halves come from one source population, so this is a boundary/sanity check rather than validation with distinct strength and stress populations. Non-rejection of the Weibull tests does not change that distinction.

Table 11: Parameter and reliability estimates with 95 percent reliability intervals. Interval types are explained in the text.

Data	Estimator	α	a	b	R	95 percent interval
Jute	MLE	1.166	2.117	4.846	0.753	[0.444, 0.945]
Jute	Ordered MLE	1.166	2.117	4.846	0.753	[0.500, 0.961]
Jute	Bayes	1.179	2.201	3.901	0.666	[0.380, 0.895]
Jute	Ordered Bayes	1.165	2.104	4.127	0.699	[0.516, 0.899]
Carbon	MLE	2.874	1.341	0.695	0.316	[0.133, 0.615]
Carbon	Ordered MLE	2.660	1.135	1.135	0.500	[0.500, 0.862]
Carbon	Bayes	2.784	1.317	0.827	0.362	[0.159, 0.608]
Carbon	Ordered Bayes	2.552	1.080	1.338	0.565	[0.502, 0.704]

Table 12: Numerical 95 percent HPD intervals for reliability in the data examples. The probability column checks mass under the interpolated posterior density.

Data	Posterior	HPD interval	Width	Probability
Jute	Unrestricted	[0.4040, 0.9115]	0.5075	0.950
Jute	Ordered	[0.5000, 0.8744]	0.3744	0.950
Carbon	Unrestricted	[0.1462, 0.5915]	0.4454	0.950
Carbon	Ordered	[0.5000, 0.6752]	0.1752	0.950

Table 11 uses a log-ratio asymptotic interval for the unrestricted MLE, a 1999-resample percentile bootstrap interval for the ordered MLE, and equal-tail intervals for the Bayesian rows. Table 12 gives the numerical HPD alternatives for the same posterior distributions. Confidence intervals and posterior credible intervals have different interpretations.

The jute estimates illustrate the calculation, but the rejected fit in one group must remain part of their interpretation. The carbon results are also sensitive to imposing an order: the ordered model excludes every reliability below one half. A narrower interval in that model is partly a consequence of the assumption, rather than evidence that the assumption is correct.

The random carbon split has likelihood diagnostic $p = 0.101$, so it is not a 5 percent rejection of the order. Its unrestricted posterior order probability, about 0.127, nevertheless gives little support to imposing it. Ordered results should be read as conditional sensitivity fits. Section 4.9 explains both measures. The primary gauge example in that section uses separate laboratory populations.

Table 13: Evidence about the rate order from retained data. The likelihood diagnostic uses 999 boundary-bootstrap samples. Posterior probability is calculated under the unrestricted primary prior.

Data	LR	Boundary p	$P(a \leq b \mid D)$
Jute	0.000	1.000	0.878
Split check	1.765	0.101	0.127
20 vs 80 mm	0.000	1.000	0.996

4.2.4 Interpreting the fitted estimates

For jute, unrestricted MLE reliability is about 0.7526 and the posterior mean about 0.6656. Bayes averages reliability over the posterior, rather than evaluating it at the mean rates. The ordered posterior mean is about 0.6989, partly reflecting exclusion of reliability below one half. For the carbon split, the unrestricted and ordered posterior means are about 0.3623 and 0.5653. That difference is sensitivity to an assumption, rather than automatic improvement. Only thirteen failures and six stress records enter the jute likelihood, so the wide uncertainty should be retained. Parameter posterior histograms and chain traces remain in the executed notebook; the report gives their numerical diagnostics.

4.3 Checking the calculations

There are two different questions here: whether the numerical calculations are stable, and whether the Weibull model fits the measurements. Stable calculations do not make a poorly fitting model true. We therefore keep the numerical checks separate from the goodness-of-fit results.

We checked the reliability product against direct integration of the conditional binomial probability and against complete-system simulation. The implementation of the observed information was compared with a finite-difference Hessian. The ordered profile was also compared with a separate constrained optimization. These checks reduce the chance of a coding or algebra error in the main estimates.

Quadrature checks compare the selected numerical grid with a finer grid and examine the normalized mass near the shape integration limits. The data applications use a finer grid than the main simulation because there are few examples to compute. Numerical HPD probability mass is checked against 0.95. Close agreement under grid refinement checks the numerical approximation used here; it is not a statement that all model assumptions are correct.

Table 14: Numerical comparisons used to check the calculation. Probability-mass checks refer to the interpolated density used for numerical HPD intervals.

Comparison	Largest difference	Scope
Reliability integral versus product	4.22×10^{-15}	Same theoretical target
Direct-system simulation versus product	0.00196	Within the stated Monte Carlo checks
Simulation reliability means, finer grid	1.37×10^{-3}	One dataset per scenario and support
Jute means, finer grid	1.38×10^{-14}	Both posterior supports
Jute equal-tail endpoints	4.65×10^{-5}	Both posterior supports
Jute HPD endpoints	3.72×10^{-8}	Both posterior supports
HPD posterior probability versus 0.95	6.58×10^{-12}	All saved Bayesian simulation fits

The independent posterior sampler uses four chains per model, with 15,000 iterations each and the first 4,000 removed as warm-up. Proposals are made in transformed coordinates so that positive parameters and the order support are respected. Conditional draws of the total rate use its Gamma distribution. Trace plots and the full retained chains are saved with the notebook.

Table 15: Checks from four retained chains for each model. Difference means sampler mean minus quadrature mean. R-hat is the classical split statistic.

Model	Target	R-hat	ESS	Mean	MCSE	Difference
Unrestricted	α	1.0006	5645	1.1722	0.0029	-0.0066
Unrestricted	a	1.0002	9428	2.1893	0.0076	-0.0120
Unrestricted	b	1.0009	8607	3.8709	0.0159	-0.0302
Unrestricted	R	1.0005	5996	0.6650	0.0017	-0.0006
Ordered	α	1.0001	6700	1.1635	0.0026	-0.0019
Ordered	a	1.0001	10088	2.0967	0.0067	-0.0072
Ordered	b	1.0002	7673	4.1293	0.0162	0.0021
Ordered	R	1.0007	3977	0.7003	0.0017	0.0014

A trace without a long visible drift is a useful check, but it is not proof of convergence. We also compare between-chain and within-chain variation through split R-hat, and report effective sample size and Monte Carlo error. Here the largest split R-hat is about 1.0009 and the smallest effective sample size is about 3977. The sampler means agree with quadrature within the stated Monte Carlo checks. These are classical split diagnostics rather than rank-normalized versions.

4.4 Sensitivity and posterior prediction

A single fitted estimate does not show how much the answer depends on modelling choices. We vary one stated feature at a time in the jute example. Table 16 distinguishes changes to the prior, the hybrid stopping scheme and the record design. Ordered and unrestricted rows must still be read as different assumptions about the populations.

Table 16: Jute sensitivity calculations. All intervals in this table are posterior equal-tail intervals; each row changes the stated input.

Change	Setting	Ordered	R	95 percent interval
Prior	Primary	No	0.6656	[0.3805, 0.8947]
Prior	Primary	Yes	0.6989	[0.5157, 0.8988]
Prior	Broader total rate	No	0.6757	[0.3886, 0.9018]
Prior	Broader total rate	Yes	0.7063	[0.5173, 0.9053]
Prior	More concentrated rates	No	0.6194	[0.3639, 0.8474]
Prior	More concentrated rates	Yes	0.6609	[0.5103, 0.8546]
Hybrid scheme	Type I	No	0.6656	[0.3805, 0.8947]
Hybrid scheme	Type I	Yes	0.6989	[0.5157, 0.8988]
Hybrid scheme	Adaptive II	No	0.5450	[0.2924, 0.7863]
Hybrid scheme	Adaptive II	Yes	0.6223	[0.5057, 0.8047]
Record design	k=1, r=4	No	0.6782	[0.3564, 0.9183]
Record design	k=2, r=4	No	0.6454	[0.3321, 0.8949]
Record design	k=2, r=8	No	0.7116	[0.4479, 0.9126]
Record design	k=3, r=6	No	0.5856	[0.3110, 0.8393]

Under the three priors, the unrestricted posterior mean reliability ranges from about 0.6194 to 0.6757. The more concentrated rate prior pulls the estimate down compared with the primary prior. This is why the primary prior is described as a fixed proper prior, rather than as noninformative.

The result should not be presented as if it depended on the observations alone.

Changing the strength protocol from Type I to adaptive Type II changes the unrestricted posterior mean from about 0.6656 to 0.5450 in this retrospective example. The adaptive rule retains a different experiment; it is not just a different label on the same data. Similarly, different record orders and stopping counts select different stress information. Their estimates need not change monotonically. In particular, retaining more records changes both the information and the largest observed record.

The original printed order is only an imposed acquisition order. We therefore repeat record extraction after 200 random permutations of the complete comparison sample. Some permutations do not produce six records in the available sequence. The plotted distribution uses successful permutations only. Its spread shows acquisition-order sensitivity; it is not a confidence distribution for a new independent sample.

Of 200 attempted permutations, 123 reach the prescribed six upper 2-records and 77 do not. Among successful orders, posterior mean reliability ranges from 0.6025 to 0.8186, with mean 0.6587 and standard deviation 0.0356. The original printed-order mean is 0.6656.

For a posterior predictive check, we draw parameters from the fitted unrestricted posterior and generate a fresh experiment using the original censoring and record rules. We repeat this 2000 times and compare the retained failure count and largest stress record with the observed values. The acquisition-order figure marks the original printed-order estimate.

Table 17: Posterior predictive summaries from 2000 regenerated censored/record experiments. The last column is a descriptive upper-tail probability.

Statistic	Observed	95 percent predictive range	Upper tail
Observed failure count	13.0000	[8.0000, 19.0000]	0.666
Largest record	0.6627	[0.2974, 2.2573]	0.651

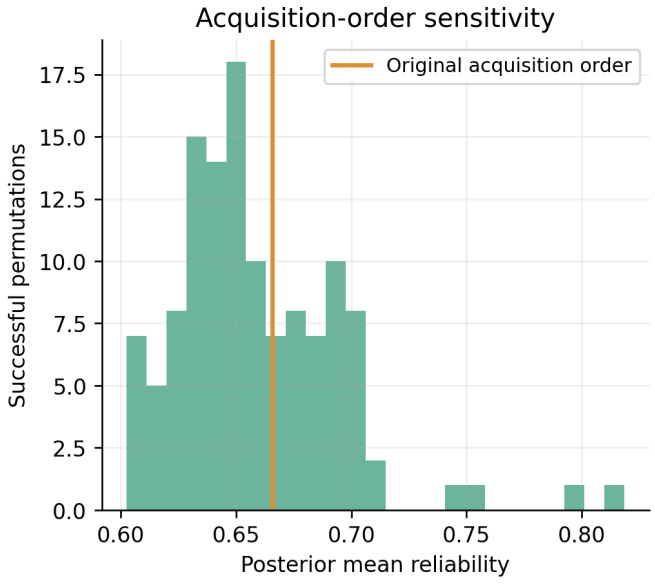


Figure 8: Posterior mean reliability after successful random acquisition-order permutations. The vertical line marks the original printed-order result.

The observed failure count, 13, falls within the predictive central range of 8 to 19. The largest retained stress record also falls within its predictive range. This says that these two summaries are not unusual under the fitted experiment. It does not cancel the complete-data jute goodness-

of-fit rejection. Posterior predictive tail areas are descriptive model checks; they are not ordinary uniformly distributed significance-test p-values.

4.5 Additional bootstrap comparisons

The main study gives an asymptotic interval for the unrestricted MLE. With few stress records, its normal approximation can be inaccurate. We therefore compare it with a percentile parametric bootstrap interval and a studentized interval. These additions address interval accuracy; they do not change the likelihood or turn the fitted data into a new independent experiment.

In the enlarged main study, the smallest unrestricted ACI coverage is 0.916 in scenario 18, with 20 specimens per block and MCSE 0.0088. A difference of a few coverage points must still be compared with its Monte Carlo uncertainty.

Let $\widehat{\psi} = \log(\widehat{b}/\widehat{a})$ and let $\widehat{\sigma}_\psi$ be its observed-information standard error. For each bootstrap dataset, refit the unrestricted model and calculate

$$T^* = \frac{\widehat{\psi}^* - \widehat{\psi}}{\widehat{\sigma}^*_\psi}. \tag{32}$$

If $t^*_{.025}$ and $t^*_{.975}$ are the empirical pivot quantiles, invert them to obtain

$$\left[\widehat{\psi} - t^*_{.975}\widehat{\sigma}_\psi, \quad \widehat{\psi} - t^*_{.025}\widehat{\sigma}_\psi\right]. \tag{33}$$

Exponentiating and applying $R(1, \rho)$ gives the reliability interval. We call this Boot-t on the log-rate ratio. Studentizing the rate ratio on its logarithmic scale keeps positive ratios and valid reliability endpoints. It is different from studentizing reliability directly. Boot-p is the interval between the empirical 0.025 and 0.975 quantiles of the bootstrap reliability estimates.

Every replicate preserves the observation design: starting block sizes, targets, time thresholds, planned withdrawals, protocol, record order and stopping count. The bootstrap is generated under the fitted parameters; the time thresholds stay at their design values rather than being recalculated from each new fit. The ordered percentile bootstrap is generated and refitted under the ordered model. We do not apply the ordinary studentized construction to a pooled ordered MLE, where the regular normal approximation fails.

Algorithm 3 Parametric bootstrap intervals for one retained dataset

Input: Retained observations, fixed observation design, bootstrap count B

1: Fit the unrestricted MLE and calculate $\widehat{\psi}, \widehat{\sigma}_\psi$
2: **for** $h = 1, \ldots, B$ **do**
3: Generate independent Weibull strength blocks under the fitted model
4: Apply each block's withdrawals and hybrid stopping rule
5: Generate stress upper k-records to the prescribed record count
6: Refit and save $\widehat{R}^*_h$ and the pivot in (32)
7: **end for**
8: Use reliability quantiles for Boot-p and inverted pivot quantiles for Boot-t
9: Repeat generation and fitting under the ordered model for ordered Boot-p

Output: Three bootstrap reliability intervals, with the original ACI as a comparator

The enlarged coverage experiment has twelve settings and 500 independent outer datasets in each setting. Each outer fit uses 399 bootstrap resamples. The settings cross three rate regimes, two block sizes and two hybrid protocols. Shape is 1.8, record order is one, and the record counts are six or twelve as in the main size contrast. The setting-level coverage and width results are saved

with the data. For coverage near 0.95, 500 replications give MCSE about 0.010; differences of one percentage point are therefore not decisive by themselves.

The full setting-by-setting bootstrap table is saved in `data/results/bootstrap_extended_summary.csv`; Table 8 reports ordered coverage and Figure 9 compares all four interval methods.

Across the four ordered-regime settings, average coverage is about 0.933 for ACI, 0.930 for unrestricted Boot-p, 0.944 for Boot-t and 0.949 for ordered Boot-p. At the equality boundary the corresponding averages are about 0.937, 0.932, 0.947 and 0.962. The studentized interval improves average unrestricted coverage in these comparisons, usually with a greater width. This is evidence from the stated settings, not a guarantee of improvement for every sample size or parameter value. Under the violated order, ordered Boot-p has zero coverage because the true reliability lies outside its support. Increasing the bootstrap count cannot fix a false support assumption.

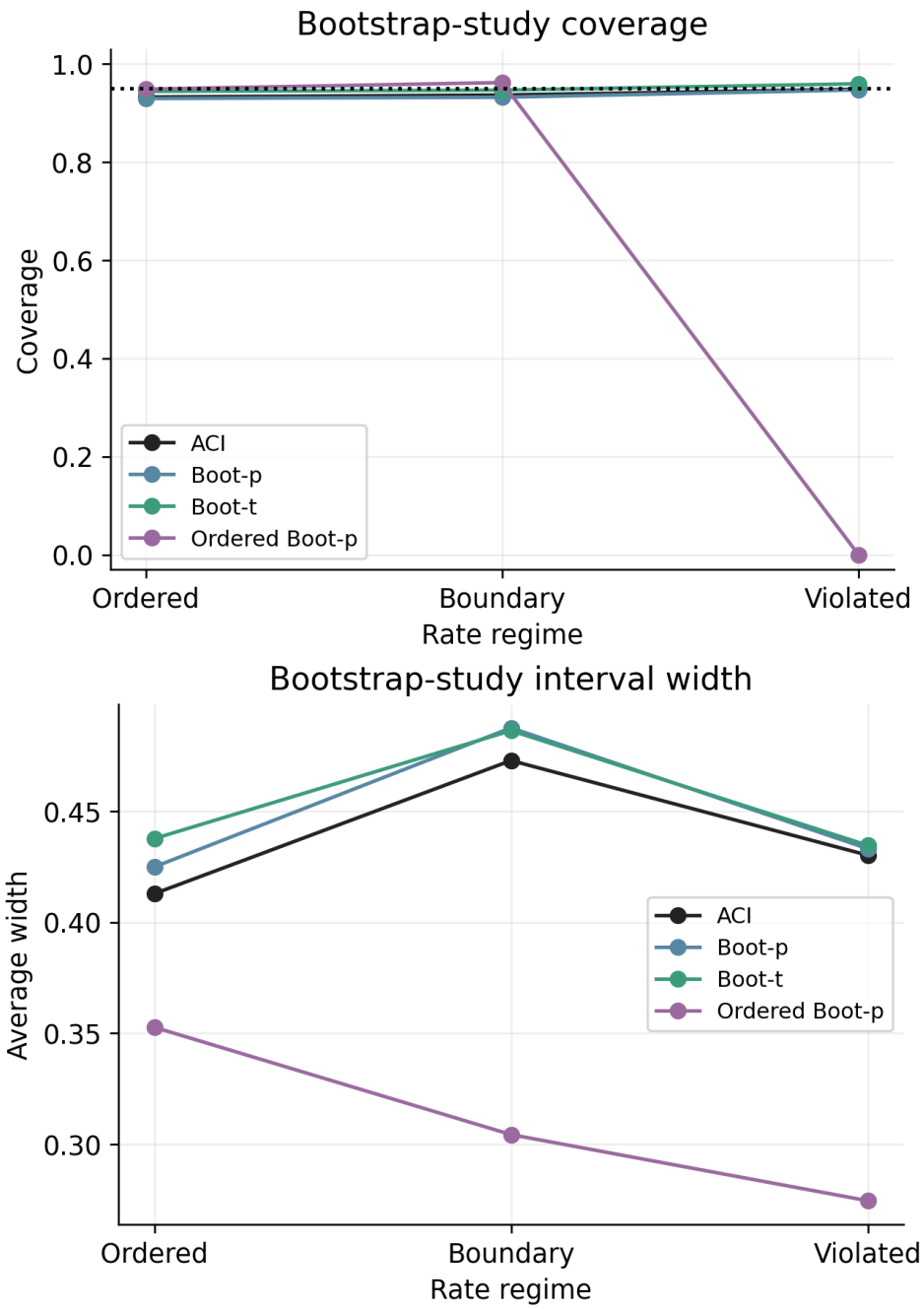


Figure 9: Coverage and average width in the expanded bootstrap study, averaged within each rate regime. The dotted line marks nominal coverage 0.95.

4.6 Another system and different observation designs

A formula may allow many s and K values while a numerical study examines only one. We add a 2-out-of-4 comparison with shape 1.8, strength rate 0.8 and stress rate 1.6. Its true shared-stress reliability is 0.8 and its ordered lower bound is 0.6. Four estimators are fitted to the same generated data in each replication. There are 500 replications in each of fourteen designs.

Eight designs cross strength block sizes 20 and 40 with stress record counts six and twelve under both protocols. This separates the two size changes that were combined in the main study. The uniform blocks use a 70 percent failure target, balanced withdrawals and the true strength 65th percentile as the fixed threshold. Record order is one. The total starting strength sample is three times the displayed block size.

Two more designs have unequal facilities. Their (n_i, m_i) values are $(15, 9)$, $(20, 14)$ and $(25, 19)$. The thresholds are the true strength 45th, 65th and 80th percentiles, and the withdrawal patterns are early, balanced and late, respectively. There are 60 starting specimens and 42 target failures in total. Early withdrawals remove the planned surplus at the first failure; late withdrawals remove it at the last failure. Actual removals still follow the chosen hybrid rule. These designs vary observation settings across facilities, while retaining one common strength population. Giving each facility a different population rate would require a different statistical model.

The last four designs use uniform blocks of size 20, six records, and either early or late withdrawals under each protocol. Table 18 gives reliability MSE and Bayesian HPD coverage for all fourteen designs. I denotes Type I and A denotes adaptive Type II.

Table 18: Additional 2-out-of-4 designs, with 500 replications each. Columns U and O denote unrestricted and ordered estimators. CP columns refer to numerical HPD intervals.

ID	Design	Rule	r	U MLE	O MLE	U Bayes	O Bayes	U CP	O CP
1	n20	I	6	0.0153	0.0119	0.0114	0.0039	0.958	0.966
2	n20	I	12	0.0138	0.0106	0.0107	0.0035	0.966	0.976
3	n40	I	6	0.0123	0.0098	0.0099	0.0036	0.964	0.964
4	n40	I	12	0.0088	0.0080	0.0075	0.0034	0.956	0.954
5	heterogeneous	I	6	0.0157	0.0120	0.0120	0.0040	0.940	0.952
6	early	I	6	0.0153	0.0115	0.0116	0.0038	0.946	0.960
7	late	I	6	0.0155	0.0121	0.0120	0.0040	0.950	0.956
8	n20	A	6	0.0110	0.0093	0.0088	0.0036	0.960	0.960
9	n20	A	12	0.0093	0.0080	0.0078	0.0036	0.952	0.952
10	n40	A	6	0.0099	0.0091	0.0077	0.0038	0.950	0.948
11	n40	A	12	0.0068	0.0065	0.0059	0.0035	0.950	0.936
12	heterogeneous	A	6	0.0118	0.0102	0.0094	0.0038	0.968	0.972
13	early	A	6	0.0111	0.0100	0.0085	0.0037	0.972	0.976
14	late	A	6	0.0144	0.0117	0.0113	0.0041	0.958	0.956

In the Type-I crossed comparison, unrestricted MLE MSE falls from about 0.0153 to 0.0138 when record count increases from six to twelve at block size 20. At block size 40 the corresponding values are about 0.0123 and 0.0088. Increasing strength size at a fixed record count also reduces MSE here. Thus the main study's size improvement cannot be assigned entirely to either source of information.

The mixed-block and withdrawal results also show that an observation design matters. For example, adaptive late withdrawals give a different unrestricted Bayesian MSE from adaptive early withdrawals. The ordered Bayesian estimator benefits from the correct order in this experiment. Its low MSE is conditional on that order and on the stated proper prior. HPD coverage varies across settings, with Monte Carlo uncertainty from 500 replications.

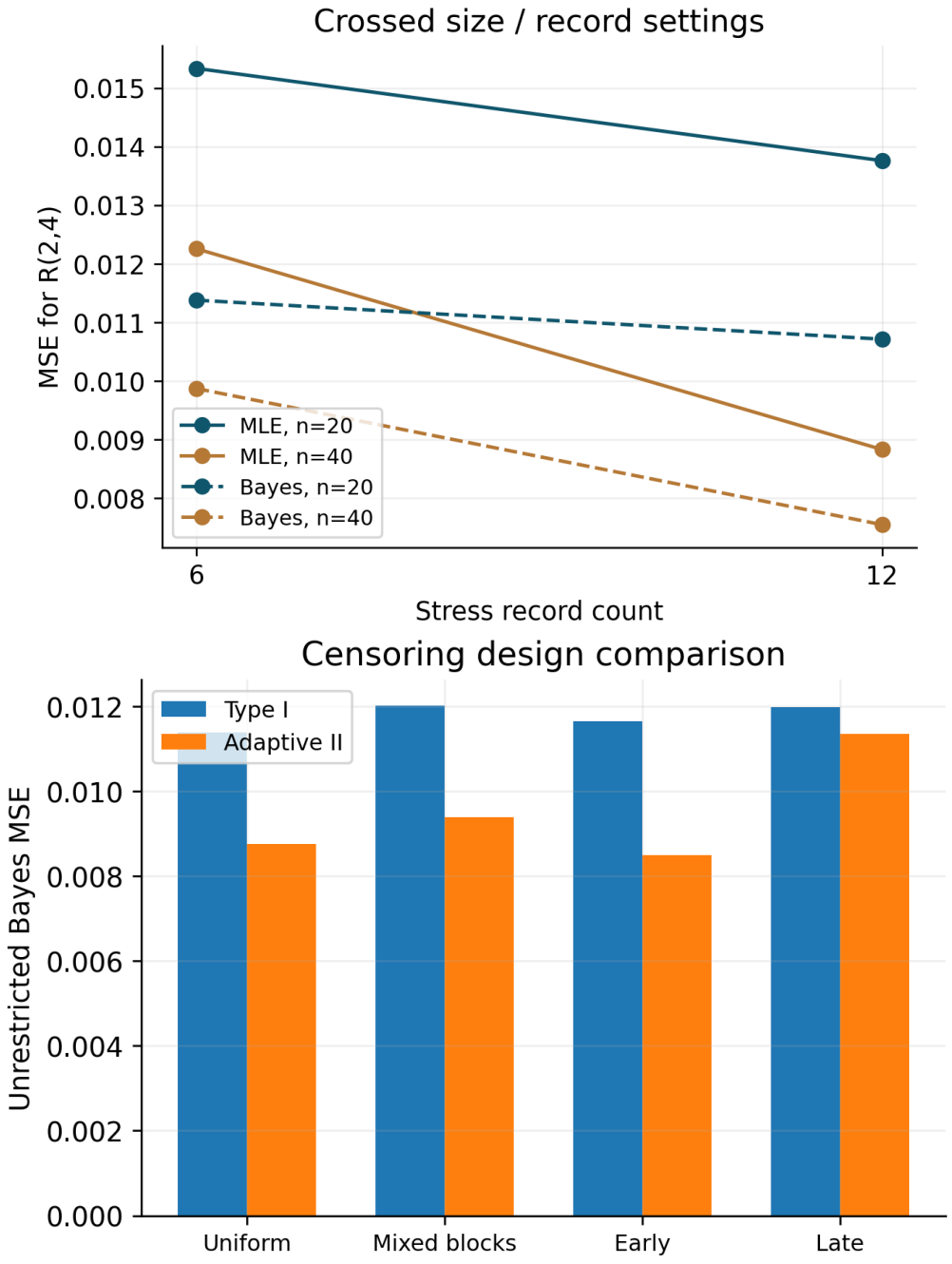


Figure 10: The added 2-out-of-4 study. The first panel crosses strength block size and stress record count under Type I; the second compares block and withdrawal designs. Each plotted setting uses 500 datasets.

4.7 How much the prior changes the simulation

The primary prior puts real information into a small retained sample. It should not be described as noninformative, and an estimator's lower MSE under one prior is not a prior-free result. To examine this, we compare the primary prior with a broader proper prior using the same 500 datasets per regime. Shape is 1.8, blocks have size 20, there are six upper 1-records, and the strength protocol is Type I.

The broader hyperparameters in (19) are

$$(a_0, b_0, c_1, c_2, a_1, b_1) = (0.5, 0.125, 1, 1, 0.5, 0.25). \tag{34}$$

Both priors have $E(t) = 4$, $E(\alpha) = 2$ and $E(p) = 1/2$. Under the primary prior, $\mathrm{Var}(t) = 8$, $\mathrm{Var}(\alpha) = 2$ and $\mathrm{Var}(p) = 0.05$. The broader prior gives variances 32, 8 and 1/12, respectively. It is still a proper prior and remains dependent in the two rates. The comparison changes concentration while keeping these central values fixed; it does not establish invariance to all reasonable priors.

Table 19: Prior comparison in 500 paired datasets per regime, with 20 specimens per block and six records. Intervals are numerical HPD intervals.

Regime	Prior	Ordered	Bias	MSE	CP	Width
Ordered	Primary	No	-0.0386	0.0138	0.942	0.458
Ordered	Primary	Yes	0.0007	0.0055	0.966	0.341
Ordered	Broader proper	No	-0.0402	0.0167	0.932	0.477
Ordered	Broader proper	Yes	0.0064	0.0064	0.962	0.346
Boundary	Primary	No	0.0224	0.0160	0.934	0.506
Boundary	Primary	Yes	0.1351	0.0211	0.888	0.286
Boundary	Broader proper	No	0.0098	0.0189	0.922	0.526
Boundary	Broader proper	Yes	0.1383	0.0224	0.890	0.293
Violated	Primary	No	0.0609	0.0162	0.942	0.478
Violated	Primary	Yes	0.2793	0.0790	0.000	0.219
Violated	Broader proper	No	0.0418	0.0166	0.936	0.487
Violated	Broader proper	Yes	0.2825	0.0810	0.000	0.227

In the ordered regime, unrestricted Bayesian MSE is about 0.0138 with the primary prior and 0.0167 with the broader prior. Ordered Bayesian MSE changes from about 0.0055 to 0.0064. Other rows show smaller or different changes. Under a false order, both priors still give large ordered bias because neither permits reliability below the ordered bound. Changing prior concentration cannot repair that support error.

The point errors, coverage and widths should be read together. A more concentrated prior can reduce MSE near values it favours, while changing bias or interval coverage elsewhere. These results support reporting the actual hyperparameters and sensitivity results alongside the main comparison.

4.8 Separate primary carbon gauge populations

The random carbon split is useful for checking behaviour near a population boundary, but it does not provide two distinct source populations. We therefore add the 20 mm and 80 mm gauge groups from the authors' public carbon-fibre workbooks [6]. The related data article is Kumar et al. [7]. Gauge length distinguishes these laboratory groups before any fitted rate is calculated. We chose the endpoint gauges before fitting, rather than searching gauge pairs for an attractive result.

The variable used is maximum failure stress in MPa. We convert it to GPa by dividing by 1000, then use a fixed 5 GPa analysis unit for both groups. The saved CSV retains every positive

measurement with an identified specimen row, giving 153 rows per gauge. The source workbook checksums match the authors' published checksums. The article describes approximately 150 fibres per gauge and reports exclusions for experimental errors. Our transparent workbook-row rule is not a reconstruction of its plotting exclusions or an exact replication of its published fitted parameters. Low measured strengths remain in our data.

The source experiment also has a selection limitation: some fibres can break during preparation before tensile testing. Those unmeasured values are absent from the published measurements. Our retrospective censoring is imposed after observing the available workbook specimens; it cannot recover missing preparation failures. The analysis describes the saved laboratory samples, conditional on their collection process.

Table 20: Separate Weibull fits for all 153 valid source rows in each primary gauge group, using the fixed 5 GPa unit. K-S p-values use 1000 refitted bootstrap samples.

Gauge	n	α	Rate	K-S	p
20 mm	153	6.128	1.933	0.0504	0.455
80 mm	153	6.156	5.569	0.0586	0.230

The separate Weibull K-S bootstrap p-values are about 0.455 and 0.230. The common-shape likelihood-ratio statistic is 0.00264, with bootstrap p-value about 0.967 from 1000 refitted samples. We therefore do not reject these working assumptions at 5 percent. This is compatibility with the observed samples, rather than proof of a Weibull law. The complete-data common shape is about 6.14, and the larger stress rate for the 80 mm group corresponds to smaller breaking strengths on the common unit scale.

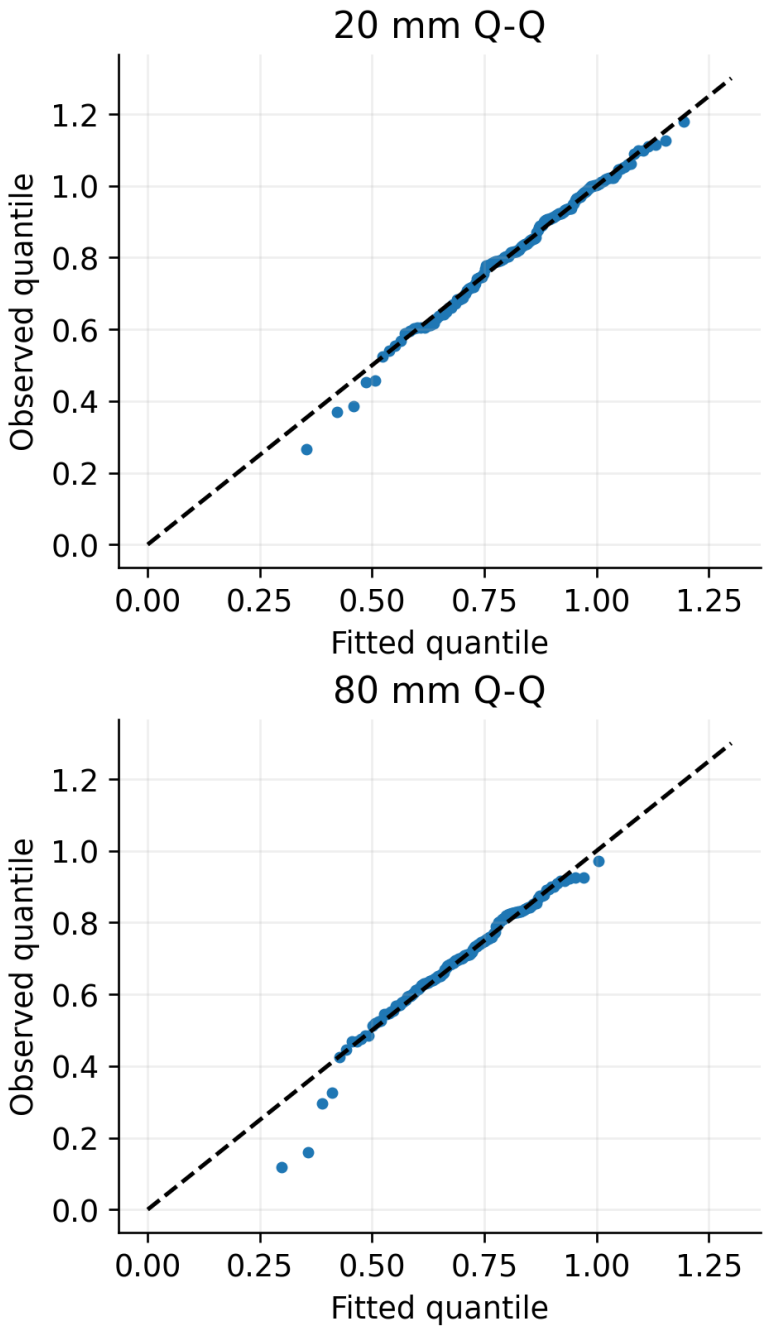


Figure 11: Primary carbon Q-Q plots for the separately measured 20 mm and 80 mm gauge groups. Density and CDF plots remain in the notebook and graphs folder.

For the retained-data illustration, the 20 mm group supplies three strength blocks of 51 specimens, with target 35 failures per block, balanced withdrawals and threshold one in the scaled unit. The 80 mm group supplies six upper 2-records. The original workbook order did not reach six such records. We therefore use one fixed seeded permutation, with seed $250150045 + 201$, and save both acquisition and original source positions. There is no repeated search for a permutation with a preferred estimate. This remains an imposed retrospective acquisition design.

Table 21: Retained-data estimates for the distinct 20 mm and 80 mm primary carbon groups. Units are 5 GPa. Bayesian intervals are equal-tail; the ordered MLE interval is percentile bootstrap.

Estimator	α	a	b	R	95 percent interval
MLE	5.830	1.828	13.345	0.9581	[0.8334, 0.9934]
Ordered MLE	5.830	1.828	13.345	0.9581	[0.8633, 0.9963]
Bayes	5.549	1.782	6.576	0.8381	[0.6184, 0.9570]
Ordered Bayes	5.549	1.781	6.594	0.8395	[0.6274, 0.9570]

The retained-data unrestricted MLE is about 0.9581, whereas the unrestricted posterior mean is about 0.8381. The ordered posterior mean is about 0.8395, very close to its unrestricted counterpart. Only six stress records enter this likelihood, even though 153 complete stress measurements are available for separate fit checks. The rate estimate can therefore be sensitive to the retained records and to the proper prior. Its high MLE should not be read as an observed population truth. The intervals show substantial uncertainty.

The primary numerical HPD intervals are $[0.6606, 0.9726]$ without the order and $[0.6652, 0.9721]$ with it. They differ from the equal-tail intervals in Table 21.

The additional bootstrap intervals for all three examples are in Table 22.

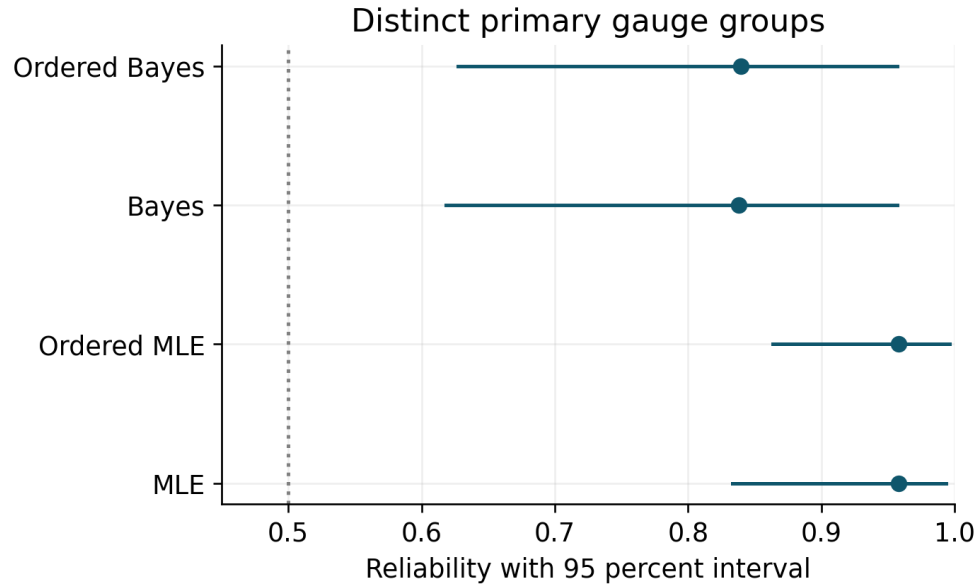


Figure 12: Retained-data reliability estimates for the distinct primary gauge groups. The Bayesian lines are equal-tail intervals and the ordered MLE line is percentile bootstrap.

4.9 Evidence about the order assumption

A restricted model is a conditional analysis, so its narrower support needs a scientific reason. As a descriptive likelihood diagnostic, compare the unrestricted and ordered maxima through

$$D_O = 2\{\ell(\widehat{\alpha}, \widehat{a}, \widehat{b}) - \ell(\widetilde{\alpha}, \widetilde{a}, \widetilde{b})\}. \tag{35}$$

It is zero if the unrestricted estimate satisfies the order. To calibrate it, fit the equality boundary $a = b$, regenerate the full experiment under that boundary fit, and refit both models in 999 replicates. The reported tail area uses $(1 + \#\{D_O^* \geq D_O\})/1000$. This is a parametric boundary calibration with fitted nuisance parameters. It is not an exact finite-sample test over the entire composite order null.

We also calculate $P(a \leq b \mid D)$ under the unrestricted primary prior by integrating its posterior over $p \leq 1/2$. It answers a Bayesian question under that prior, rather than the repeated-sampling question of the likelihood diagnostic. Table 13, placed beside the first data estimates, reports both measures.

For the jute example, posterior order probability is about 0.878, but the complete-data Weibull rejection still limits interpretation. The random split has posterior order probability about 0.127 and boundary-bootstrap tail area 0.101. Calling this a 5 percent rejection would overstate the evidence. Its ordered analysis is a sensitivity fit, not the preferred result solely because it is narrower. The primary gauge groups have posterior order probability about 0.996. In jute and primary examples the unrestricted MLE already satisfies the order, so the likelihood diagnostic is zero and its tail area is one. A tail area of one does not mean the order has probability one.

These diagnostics do not define a new automatic method that tests, selects an estimator and then reuses its interval with guaranteed coverage. The report keeps both fits visible so that the assumption and its effect can be assessed.

Table 22: Additional application intervals. Boot-t studentizes the unrestricted log-rate ratio and transforms endpoints to reliability. O-Boot-p is ordered MLE percentile bootstrap.

Data	Interval	Estimate	Lower	Upper	B
Jute	ACI	0.7526	0.4445	0.9446	0
Jute	Boot-p	0.7526	0.4200	0.9569	3999
Jute	Boot-t	0.7526	0.4001	0.9433	3999
Jute	O-Boot-p	0.7526	0.5000	0.9587	3999
Split check	ACI	0.3160	0.1327	0.6146	0
Split check	Boot-p	0.3160	0.1259	0.6655	3999
Split check	Boot-t	0.3160	0.1063	0.5999	3999
Split check	O-Boot-p	0.5000	0.5000	0.8474	3999
20 vs 80 mm	ACI	0.9581	0.8334	0.9934	0
20 vs 80 mm	Boot-p	0.9581	0.8665	0.9966	3999
20 vs 80 mm	Boot-t	0.9581	0.7817	0.9905	3999
20 vs 80 mm	O-Boot-p	0.9581	0.8633	0.9963	3999

4.10 A complete-data comparison without the Weibull assumption

The jute rejection motivates an estimate that does not require a Weibull distribution or equal shapes. Let N_X and N_Y be complete sample sizes, and let $M_l = \sum_i \mathbf{1}\{X_i > Y_l\}$ be the number of strengths above stress observation Y_l. An empirical shared-stress estimator is

$$\widehat{R}_U = \frac{1}{N_Y \binom{N_X}{K}} \sum_{l=1}^{N_Y} \sum_{j=s}^{K} \binom{M_l}{j} \binom{N_X - M_l}{K - j}. \tag{36}$$

A binomial coefficient is zero when its lower argument is outside its possible range. For a fixed observed stress, this counts all subsets of K distinct strengths that have at least s exceedances. Averaging over stress observations gives a two-sample U-statistic for the same common-stress event. Under independent identically distributed observations within each group, it is unbiased. It does not assume independent component survival after averaging over the shared stress.

We independently resample both complete groups with replacement 2000 times and report percentile intervals. These intervals are still bootstrap approximations. Independence of specimen measurements remains an assumption. The method uses complete data and is therefore not a nonparametric solution to the retained censoring/record likelihood.

Table 23: Complete-data nonparametric shared-stress estimates with 2000 two-sample bootstrap resamples. These use all measurements, unlike the retained-data likelihood.

Data	U-statistic	Weibull	NP 95 percent interval	n_X	n_Y
Jute	0.5421	0.5231	[0.3551, 0.7243]	30	30
Split check	0.3888	0.4393	[0.2456, 0.5533]	50	50
20 vs 80 mm	0.8091	0.8110	[0.7365, 0.8707]	153	153

The complete-data nonparametric estimates are about 0.542 for jute, 0.389 for the random split, and 0.809 for the primary gauge groups. The matching complete-data common-shape Weibull estimates are about 0.523, 0.439 and 0.811. The primary parametric and nonparametric estimates are close on the same information basis. This provides a useful check on the complete-data model calculation. It does not remove selection, independence or operating-load limitations.

The primary retained-data MLE is higher than its complete-data estimate. The two calculations intentionally use different observations: one retains only censored strength failures and six stress records, while the other uses all 306 measurements. Their difference should not be described as a numerical contradiction, or as evidence that the retained estimator is automatically more accurate. The same distinction applies to the jute estimates.

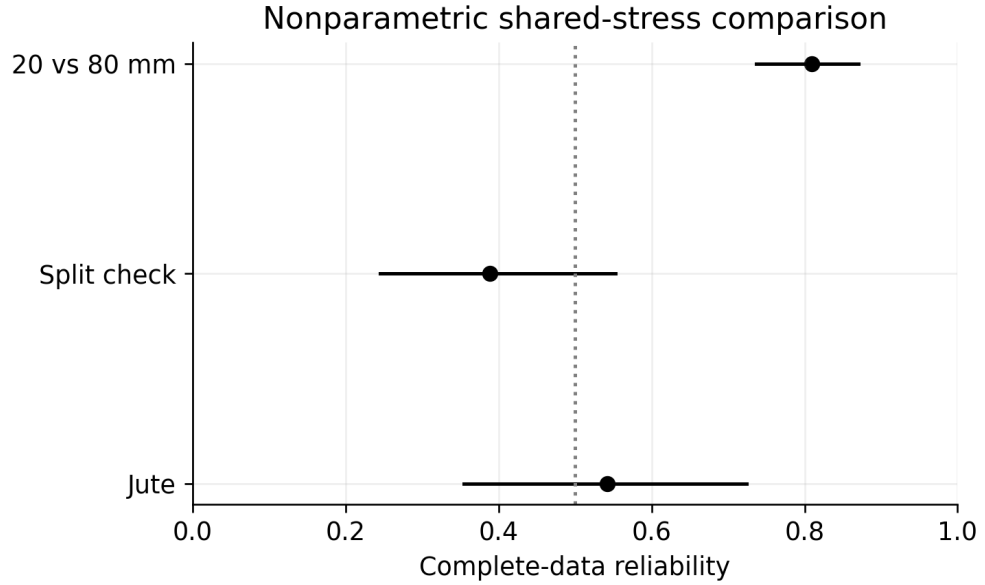


Figure 13: Complete-data nonparametric shared-stress reliability estimates and two-sample percentile bootstrap intervals. These use more observations than the retained-data fits.

4.11 Further numerical verification and scope

After integrating out the total rate t, the posterior has only two numerical coordinates. Deterministic quadrature is a practical choice in this dimension and avoids Monte Carlo noise in the repeated estimator comparison. It is an approximation: finite shape bounds, node counts and interpolation can affect it. Refinement checks and independent sampling therefore remain necessary.

Across all three data examples, changing from 240×320 to 400×520 quadrature nodes changes posterior means by at most 1.4×10^{-13}, equal-tail endpoints by less than 6.4×10^{-5} and HPD endpoints by less than 3.9×10^{-7}. Shape-edge mass is negligible in these examples. These checks examine numerical integration, not distributional fit.

We add independent four-chain comparisons in ordered, boundary and violated scenarios for the 2-out-of-4 system. Each chain runs for 10,000 iterations and discards the first 2000. Table 24 summarizes differences between quadrature means and sampler means. The largest discrepancy is about 2.2 sampler MCSEs, with the largest split R-hat near 1.0013 and the smallest effective sample size above 2800. This supports agreement in the tested cases without proving convergence for every dataset.

Table 24: Independent 2-out-of-4 checks in ordered, boundary and violated scenarios (IDs 1, 2 and 3). Four chains retain 8000 iterations each. The last column is the largest absolute mean difference divided by its sampler MCSE.

ID	Ordered	Max R-hat	Min ESS	Max MCSE units
1	No	1.0008	4165	1.14
1	Yes	1.0009	2821	0.98
2	No	1.0009	3774	2.16
2	Yes	1.0013	2960	1.31
3	No	1.0012	4041	2.05
3	Yes	1.0010	3971	1.06

The batched optimizer used in the larger bootstrap comparison is checked against 200 independent scalar fits, with largest parameter difference about 5.3×10^{-8}. The faster generator uses exponential spacings of the transformed Weibull lifetimes. Its retained-count mean is compared with the original complete-lifetime generator using 30,000 and 3000 samples, respectively. Specimen conservation is checked for each heterogeneous block. The implementation also checks the reliability formula against the conditional integral for four system choices and three rate ratios.

Several limits remain. The product reliability formula requires a common shape; with unequal shapes the conditional integral in (4) must be evaluated for the unequal-shape model. The upper-record likelihood describes record values stopped at a prescribed record count. It deliberately excludes acquisition waiting times; saving their positions does not add those times to the likelihood. The applications impose censoring and acquisition orders retrospectively, and their comparison populations are laboratory breaking strengths rather than measured operating loads. The report therefore gives a reproducible statistical comparison, not a certified engineering reliability.

5 Conclusion

We derived the shared-stress reliability of a Weibull s-out-of-K system and used the appropriate likelihood for block progressive hybrid strengths and upper k-record stresses. Both unrestricted and ordered MLEs were obtained by profiling over the common shape. A dependent gamma-Dirichlet prior gave Bayesian estimates in both cases, with numerical integration reducing the posterior calculation to two dimensions.

The simulation compares point-estimation errors and the coverage and width of asymptotic, equal-tail and numerical HPD intervals. The ordered Bayesian estimates perform well in the correctly ordered cases studied here. A false restriction produces substantial bias, and intervals constrained above one half cannot cover a reliability below one half. At the equality boundary, HPD and equal-tail intervals behave differently because an HPD interval may include the boundary itself.

Published measurements provide reproducible illustrations, rather than proof of engineering performance. The jute example has a documented fit problem; the random carbon split is a boundary check. Separate 20 mm and 80 mm carbon groups give compatible complete-data Weibull fits and a nonparametric reliability near 0.81. The common-shape assumption, noninformative withdrawals and the imposed record acquisition order are limitations. The notebook includes prior and acquisition-order sensitivity checks and posterior diagnostics. The added 2-out-of-4 experiments separate the effects of strength size and record count and compare different block and withdrawal plans. Bootstrap studentization improves some coverage results here, but none of the approximate intervals has guaranteed finite-sample coverage. Broader-prior comparisons show that small-sample Bayesian performance depends on the prior. Unequal shapes and a model for informative withdrawals remain outside the project.

A Sources and reproducibility

Sultana et al. [1] provide the block Weibull framework, order restriction and dependent prior, and reproduce the jute measurements. Newer [2] provides the combined censoring/record framework for another distribution. Newer and Alanazi [3] provide related reliability and bootstrap ideas and the reproduced carbon table. We derive the assigned Weibull model separately. The primary gauge workbooks are from the authors' Zenodo deposit [6], linked by Kumar et al. [7].

The other supplied hybrid-inference paper informs interval reporting. The supplied attention-based control-chart paper is outside this model, so its trading data and machine-learning methods are not used. For the assigned SAGE reference, this work used the publisher abstract and citation details; full text was not accessed in this session. Source-by-source details are in `code/notes/literature_review.md`.

Full measurement listings, retained observations and the 48-row simulation design are stored in `data/`, rather than repeated in the report. The executed notebook and statistical scripts are in `code/`; all additional plots remain in `graphs/`. The primary workbook filenames, row rule, checksums and units are recorded in `data/raw/primary_carbon_provenance.json`. Seeds and priors are fixed in the code. The README gives execution and compilation commands.

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